

CORNELL UNIVERSITY

AIAA DBF
Cornell University 2025-26
Design Report



Duck Force One

Variables

b	Span
c	Chord
C	Battery capacity
C_c	2 - Cargo operating cost per lap
C_D	Coefficient of drag
C_e	10 - Base operating cost per lap
C_L	Coefficient of lift
C_l	Roll moment coefficient
C_m	Pitching moment coefficient
C_n	Yaw moment coefficient
C_p	0.5 - Passenger operating cost per lap
e	Oswald efficiency factor
g	Acceleration due to gravity
I	Current
Ic_1	10 - Fixed income per cargo
Ic_2	8 - Income per cargo per lap
Ip_1	6 - Fixed income per passenger
Ip_2	2 - Income per passenger per lap
K_τ	Motor torque coefficient
K_v	Motor speed coefficient
l	Tail moment arm
n	Load factor
Q	Torque
S	Planform area
v	Velocity
V	Voltage
$\bar{v}$	Tail volume coefficient
α	Angle of attack
β	Sideslip angle
δ	Deflection
Φ	Bank angle
ζ	Damping ratio
ω	Turn rate

Subscripts

a	Aileron
e	Elevator
HT	Horizontal stabilizer
m	Motor
r	Rudder
VT	Vertical stabilizer
w	Wing

Units

A	Amps
F	Fahrenheit
ft	Feet
in	Inches
lbf	Pound force
mAh	Milliamp hours
psi	Pounds per square inch
sec	Seconds
V	Volts
W	Watts
Wh	Watt hours

Abbreviations

AIAA	American Institute of Aeronautics and Astronautics
AR	Aspect Ratio
CFD	Computational Fluid Dynamics
CFRP	Carbon Fiber Reinforced Polymer
CG	Center of Gravity
COTS	Commercial Off the Shelf
CUDBF	Cornell University Design Build Fly
DBF	Design Build Fly
DF1	Duck Force One
EF	Efficiency Factor
FEA	Finite Element Analysis
FoM	Figure of Merit
FoS	Factor of Safety
GM	Ground Mission
GMTV	Ground Mission Test Vehicle
LDPE	Low-density Polyethylene
M1	Mission 1
M2	Mission 2
M3	Mission 3
MDO	Multidisciplinary Design Optimization
MTOW	Maximum Take-off Weight
NP	Neutral Point
PCA	Principal Component Analysis
Pr	Prandtl Number
RAC	Rated Airplane Cost
Re	Reynolds Number
RPL	Rapid Prototyping Lab
SM	Static Margin
TMS	Total Mission Score

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1 Executive Summary

This report documents CUDBF's approach to the 2026 AIAA DBF competition in Wichita, Kansas. The team's objective is to design and manufacture a bush plane capable of conducting charter flights and towing a banner. The aircraft completes three flight missions: flying a timed course in M1, carrying passengers and cargo in M2, and deploying a custom banner in flight during M3. The GM focuses on the loading and unloading of the cargo and passengers, along with flight preparation.

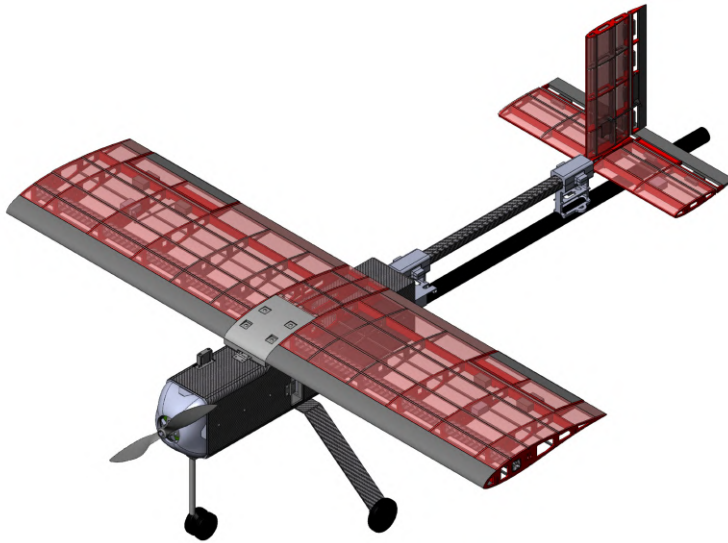


Figure 1.0.1: DF1 finalized design in M3 configuration

The team's competition entry is Duck Force One (DF1). Shown in **Figure 1.0.1**, DF1 implements a high-wing, single-motor configuration with a conventional empennage. The aircraft is capable of carrying passengers and cargo in its fuselage, with a banner mounted on the tail boom. A rigorous sizing and scoring analysis directed the team to optimize for a large number of laps in M2, a long banner in M3, and a quick GM time. DF1, therefore, carries **3** rubber duck passengers and **1** hockey puck cargo, as well as a **150.00 in** long banner with an AR of **5.00**. A **46.51 in** wingspan and **12.09 in** chord are employed to minimize stall speed and incurred RAC. Preliminary analysis detailed in **Table 1.0.1** predicts lap times of **37.10 sec** and **59.90 sec** for M2 and M3, yielding **8** and **5** laps, respectively. During M2, these parameters

predict an income of **140**, a cost of **100.66**, and an EF of **0.89**. During M3, these parameters predict an RAC of **0.94**. With minimal payload, the team also predicts a **14.00 sec** GM time. The aircraft is constructed out of carbon fiber, wood, and 3D printed components, and employs a MonoKote covering.

The remainder of this report documents CUDBF's design process for the 2026 competition cycle. **Section 2** provides an overview of the team structure, outlining the team's leadership organization, workflow, and timeline. **Sections 3** and **4** discuss the conceptual and preliminary design phases, including mission requirements, sensitivity analyses, trade studies, aircraft sizing, and aerodynamic evaluations. **Section 5** contains the detailed design of DF1's subsystems, flight performance estimates, and a CAD drawing package. **Section 6** provides a description of the manufacturing process and methodologies. Finally, **Sections 7** and **8** present a report of completed subsystem and system tests, and the results from those tests respectively.

Parameter	M1	M2	M3	GM
Takeoff Weight (lbf)	6.55	7.23	7.37	—
Cruise Speed (ft/s)	90.8	90.88	52.48	—
Number of Laps	3	8	5	—
Lap Time (sec)	35.00	37.10	59.90	—
Mission Score	1.00	1.02	2.30	0.80

Table 1.0.1: DF1 predicted mission parameters

2 Management Summary

2.1 Team Organization

CUDBF is a student-run project team at Cornell University consisting of 43 undergraduate students with 3 senior members and 40 undergrads. The team is led by a chief engineer and a project manager who are supported by a safety officer. The chief engineer is responsible for the technical progress of the team through directing the design, building, testing, and analysis. The project manager is responsible for logistics, including recruitment and scheduling. The safety officer ensures team well-being by establishing workspace safety procedures and serving as a point of contact for members. The chief engineer and project manager meet regularly with the faculty advisor for technical feedback, and once per semester with Cornell Project Teams staff to review administrative objectives, budgets, and timelines. Senior members support design development through participation in work sessions and design reviews.

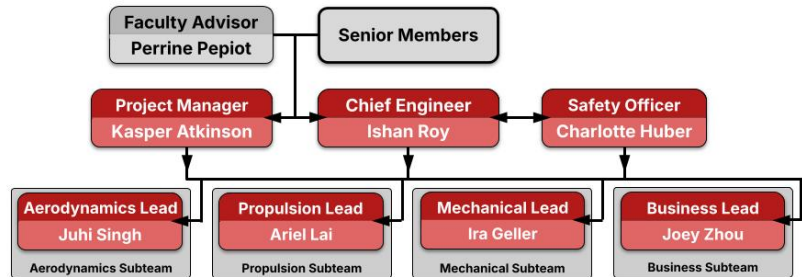


Figure 2.0.1: CUDBF's organizational structure. Subteam leads are indicated in pink

The team is organized into four subteams, distinguished by responsibility and function as shown in **Figure 2.0.1**. Weekly leads meetings foster communication between full-team leads and subteam leads. Each subteam lead directs subteam-level activities and ensures relevant subsystem requirements are met. Subteam members answer to their subteam lead, who in turn reports to the chief engineer. For cross-subteam efforts, such as system level testing, the chief engineer directs the relevant subteam leads and has final decision making authority. Both the chief engineer and project manager attend all work sessions to provide support. The role of each subteam and skills required of the members are presented in **Table 2.1.1**.

Subteam	Responsibilities	Individual Skillsets
Aerodynamics	Responsible for aircraft performance, sizing analyses, and aerodynamic surface design, including the wing and empennage.	Proficiency in aerodynamic theory, MATLAB [1], SOLIDWORKS [2], and tools such as XFLR5 and XFOIL [3].
Propulsion	Responsible for propulsion system configuration, sizing, and integration. Also responsible for avionics integration and power distribution.	Knowledge of electric propulsion systems and propeller theory, experience using tools such as eCalc [4]. Proficiency in soldering, wiring, and electronics integration.
Mechanical	Responsible for aircraft mechanisms, subsystem integration, and structural design, including the fuselage, landing gear, and mission-specific mechanisms.	Experience with mechanical design. Knowledge of statics and material mechanics with proficiency in SOLIDWORKS and analysis tools such as ANSYS [5].
Business	Manages team budget, purchases, and financial tracking. Directs team marketing, outreach, and external communications.	Budgeting experience. Strong communication skills with familiarity in web design and social media.

Table 2.1.1: Subteam responsibilities and skillsets

2.2 Timeline and Milestones

A high-level Gantt chart was created at the beginning of the academic year to track important milestones and overall progress. This schedule allocates time for three prototype aircrafts prior to the final competition vehicle, with sufficient time for testing and design revisions between each iteration. **Figure 2.2.1** shows the Gantt chart used for the 2025-2026 DBF cycle.

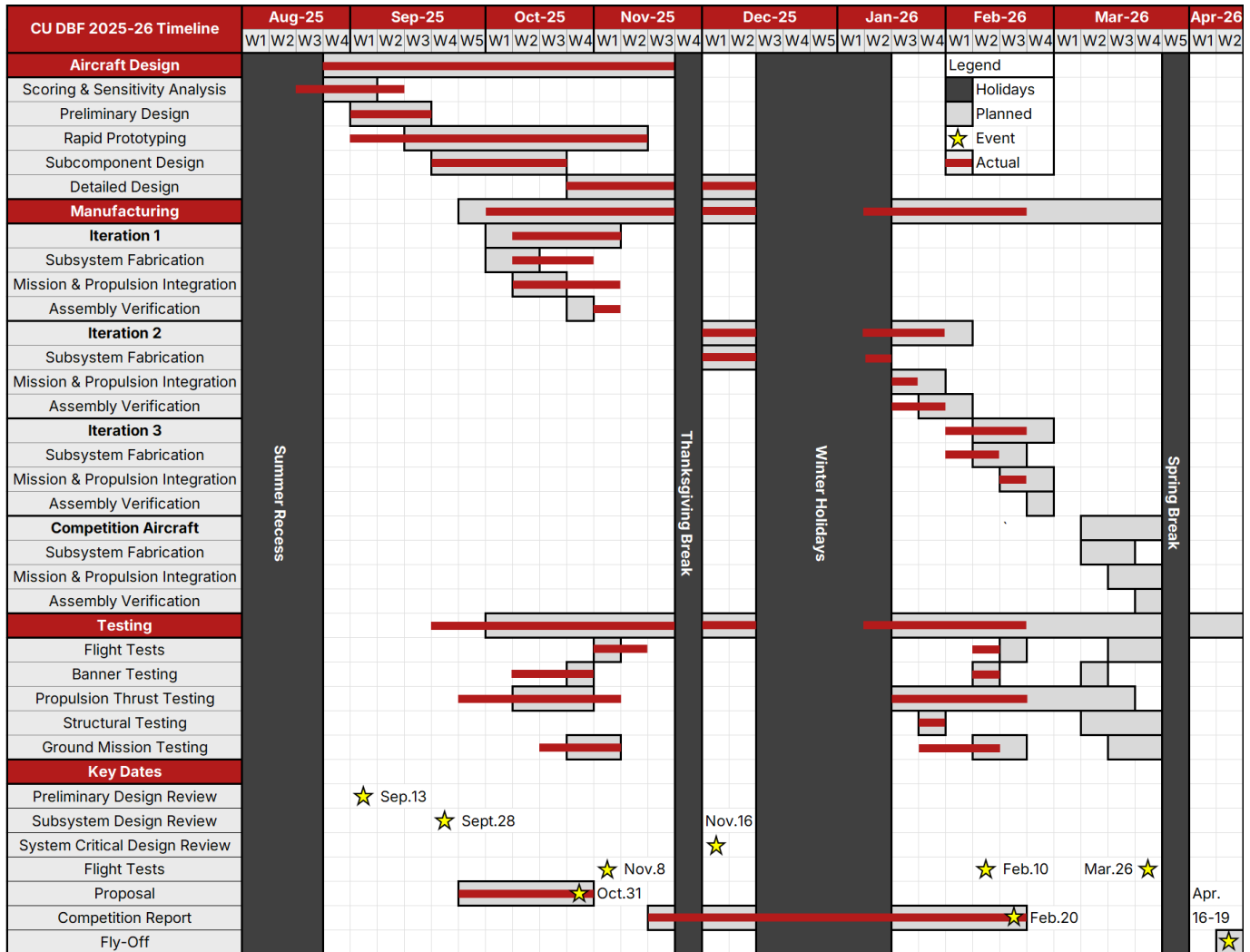


Figure 2.2.1: CU DBF Gantt chart and milestones

3 Conceptual Design Approach

3.1 Mission and Design Requirements

3.1.1 Problem Statement

The theme of the 2025-2026 AIAA DBF competition is Banner Towing Bush Plane. The missions will score the aircraft's flight capabilities (M1), passenger and cargo transportation abilities (M2), banner towing (M3), and configuration ergonomics (GM). The objective of the competition is to design, manufacture, and optimize an aircraft to perform all missions and achieve the highest mission score.

3.1.2 General Requirements

Each flight mission begins in the staging box with the aircraft in its designated flight configuration, without propulsion batteries, payloads, or arming fuse installed. Once called to the staging box, the ground crew has five minutes to install these components and any mission-specific systems to prepare the aircraft for flight. The competition consists of three sequential flight missions, as described in **Section 3.1.1**. A mission may not be attempted until a successful score has been obtained for the preceding mission. All missions use the standard competition flight pattern, consisting of a ground rolling takeoff,

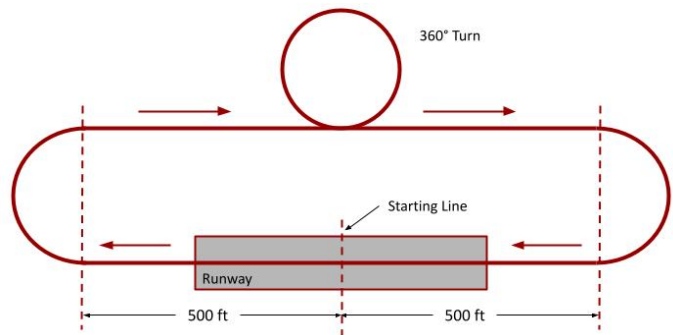


Figure 3.1.1: Competition flight pattern

multiple laps of the prescribed flight course, and a ground rolling landing on the runway, as shown in **Figure 3.1.1** [6].

Each mission has a five-minute flight window, starting when the throttle is advanced for the first takeoff attempt. A lap is counted when the aircraft crosses the start/finish line in the air. The landing does not count toward the flight window but must be successfully completed for a valid score. Flights resulting in significant damage, as determined by the flight line judge, will not receive a score. The Total Score is a function of the Design Report and Proposal scores, the combined scores from all three flight missions and GM, and participation as shown in **Equation 1**. Mission scores are normalized relative to the highest performing teams in each category. Participation points are awarded for attending the Fly-Off, passing tech inspection, and attempting a flight mission.

$$Total\ Score = (0.15 \times Proposal\ Score + 0.85 \times Design\ Report\ Score) \times (M1 + M2 + M3 + GM) + Participation \quad (1)$$

3.1.3 Mission Specific Requirements

Mission 1: Test Flight M1 is intended to demonstrate fundamental flight capability, with the aircraft flying three laps in a payload-free configuration. Successful completion of the mission in under five minutes earns one point, as shown in **Equation 2** below.

$$M1 = \begin{cases} 1 & \text{Successful mission} \\ 0 & \text{Unsuccessful mission} \end{cases} \quad (2)$$

Mission 2: Charter Flight M2 is a demonstration of the aircraft's performance while carrying a payload of passengers and cargo. The subscore given to each team is calculated as a function of the net income generated during the mission. Defined as the difference between income and cost, net income depends on the number of passengers and cargo carried, the number of laps completed within a five-minute window, and operating costs which are influenced by battery capacity. Completion of the mission earns a base score of 1 point, with an additional normalized subscore shown by **Equation 3**. Consequently, M2 has a maximum achievable score of 2 points.

$$M2 = 1 + \frac{[\text{Income} - \text{Cost}]_{\text{CUDBF}}}{[\text{Income} - \text{Cost}]_{\text{Max}}} \quad (3)$$

$$\text{Income} = N_{\text{passengers}} \times (lp1 + (lp2 \times N_{\text{laps}})) + (N_{\text{cargo}} \times (lc1 + (lc2 \times N_{\text{laps}}))) \quad (4)$$

$$\text{Cost} = N_{\text{laps}} \times (Ce + (\# N_{\text{passengers}} \times Cp) + (N_{\text{cargo}} \times Ce)) \times EF \quad (5)$$

$$EF = \frac{C \times V}{100} \quad (6)$$

Mission 3: Banner Flight M3 involves the aircraft flying as many laps as possible within the five-minute allotted time with a demonstration of the aircraft's banner towing ability. After the first upwind turn, the aircraft's banner must be remotely deployed on the down wind leg of the pattern. The banner must then be remotely released after the aircraft crosses the finish line on the final lap. The score for this mission is calculated based on the number of laps the aircraft completes, the banner length, and RAC, as shown in **Equations 7 and 8**.

$$M3 = 2 + \frac{[N_{\text{laps}} \times \frac{L_{\text{banner}}}{RAC}]_{\text{CUDBF}}}{[N_{\text{laps}} \times \frac{L_{\text{banner}}}{RAC}]_{\text{Max}}} \quad (7)$$

$$RAC = 0.05 \times \frac{b}{12} + 0.75 \quad (8)$$

GM: Aircraft Preparation The GM is a timed demonstration of the team's ability to load the aircraft's passengers, cargo, and banner. Once the timer begins, passengers enter the aircraft individually, and the cargo is loaded, after which the timer is stopped. After verifying that the passengers and cargo are secure, the timer continues, and the passengers and cargo are unloaded from the aircraft, followed by the banner being installed, after which the timer is stopped again. Lastly, while holding the aircraft vertically, the aircraft's ability to deploy and release the banner is demonstrated. The score for the GM is calculated based on **Equation 9**.

$$GM = \frac{[\text{Mission Time}]_{\text{Min}}}{[\text{Mission Time}]_{\text{CUDBF}}} \quad (9)$$

3.1.4 Mission Requirements Decomposition

This section summarizes the requirements and constraints for the four missions. Mission requirements were translated into system requirements for each subsystem to guide configuration discussions and design choices. Requirements taken directly from the rules are labeled with “-R,” while desirable requirements derived from the rules are labeled with “-D.” **Table 3.1.1** provides the full requirement breakdown, highlights subsystem tradeoffs, and helps identify optimization opportunities.

Category	Subcategory	Requirement	Number
	Configuration	Airplane's wingspan must be between 3 and 5 ft, inclusive	AC-R-01
		Aircraft MTOW must not exceed 55 lbs	AC-R-02
		Aircraft weight must be minimized	AC-D-03
		Components cannot detach from the aircraft except in M3 where the banner is released	AC-R-04
		Arming fuse must not be within 6 in from the plane of a propeller	AC-R-05
		Aircraft must have one passenger compartment and at least one cargo compartment	AC-R-06
Aircraft	Structure	Aircraft must be structurally capable of supporting the payload when subjected to the maximum G-loads of flight	AS-R-01
		All components must be adequately secured to the aircraft	AS-R-02
	Aerodynamics	The aircraft must be laterally and longitudinally stable in flight	AA-D-01
		Wings must be able to sustain steady level flight; drag must be minimized	AA-D-02
	Propulsion	Brushed or brushless electric motors must sustain flight for at least five minutes	AP-D-01
		The airplane must have an externally accessible switch to turn on the radio control system	AP-R-01
		A separate battery is required for the Rx/Servo system on the airplane	AP-R-02
		The arming fuse discharge current rating must not exceed 100A	AP-R-03
		Batteries must be COTS with a capacity under 100 Wh	AP-R-04
	1	Aircraft must fly three laps in 5 minutes with a successful landing	M1-R-01
Missions	2	Aircraft must fly as many laps as possible within 5 minutes with a successful landing	M2-R-01
		Aircraft must fly at a high velocity to maximize lap count and thus score	M2-D-01
		Payload of at least 3 passengers per cargo must be carried with a minimum of 3 passengers and 1 cargo	M2-R-02
	3	Aircraft must fly as many laps as possible within 5 minutes with a successful landing	M3-R-01
		Aircraft must be able to deploy, tow, and release a banner in flight	M3-R-02
	GM	All payloads must be able to be installed while the airplane is on its landing gear without being picked up	MG-R-01
Subsystems	Passenger Compartment	Accessed through hatch solely for access to the passenger compartment	SP-R-01
		Hatch must be ergonomic for quick loading	SP-D-02
		Passengers must be fully restrained vertically on a single horizontal plane (minimum of three)	SP-R-03
	Cargo Compartment	Accessed through hatch solely for access to the cargo compartment	SC-R-01
		Hatch must be ergonomic for quick loading	SC-D-02
		Cargo must be fully restrained for all phases of flight	SC-R-03
	Banner	Must display university name/logo on both sides	SB-R-01
		Must remain vertical in flight	SB-R-02
		Banner AR must be at most 5	SB-R-03
	Deploy/Release Mechanism	Must remotely deploy and then remotely release banner in flight	SD-R-01
		The deployment and release mechanism must be installable for GM	SD-R-02

Table 3.1.1: Aircraft design and mission requirements

3.2 Scoring Sensitivity Analysis

A scoring sensitivity analysis built around a simple flight model was performed in MATLAB [1], with results presented in **Figure 3.1.3**. Baseline best-case performances were established for each mission, assuming that at least one competing team would optimize exclusively for that mission. These reference values were derived from prior DBF competition results, with particular emphasis on the 2020 competition year due to its comparable missions. For GM and M3, baseline values of a **12 sec** GM time and a **300 in** banner length were selected. For M2, a minimum battery capacity of **2000 mAh** and a representative lap time of **35 sec** were used, along with the 3:1 passenger-to-cargo ratio and an **18 lbf** maximum payload. Using these reference conditions, the expected net income landscape, illustrated in **Figure 3.1.2**, yielded a maximum projected value of **2280**.

Score-relevant parameters derived from **Section 3.1** were varied to evaluate their influence on the TMS in **Figure 3.1.3**. As covered in **Section 4.3.6**, banners with higher ARs produce less drag [7]; therefore, the maximum allowable AR of **5.00** was used. The sensitivity analysis results indicated that GM and wingspan have the most significant effect on TMS, followed by passenger count, cargo weight, banner length, and battery capacity. The positive correlation between larger banner length and higher TMS, coupled with the negative influence of additional passengers, suggests that designs optimized for GM and M3 outperform those favoring M2 within the defined scoring framework. The design was therefore focused on minimizing passenger and cargo weight, while sizing the wingspan and shortening GM time to maximize mission performance. Additionally, it was necessary to ensure the aircraft can maintain controlled flight close to stall speeds in order to carry a longer, higher-drag banner without exceeding the capabilities of a lightweight propulsion system. The negative curvature displayed in the banner-length curve indicates the diminishing returns of excessively long banners. Overall, these insights directly informed the aircraft's subsequent sizing and configuration decisions.

3.3 Configuration Selection and Trade Studies

The conceptual design phase translated the mission requirements summarized in **Table 3.3.1** into a set of candidate aircraft configurations and identified a baseline geometry for further development. During this phase, multiple aircraft

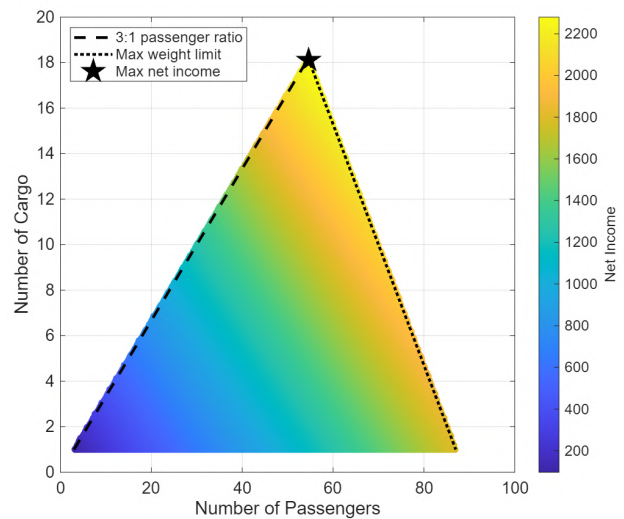


Figure 3.1.2: M2 net income as a function of payload.

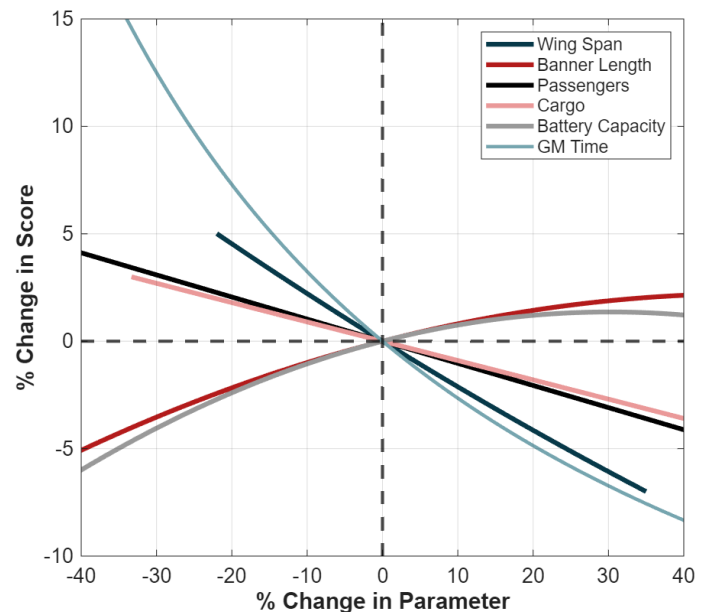


Figure 3.1.3: Parameter sensitivity study.

concepts were evaluated to determine the configuration that best satisfied mission performance, manufacturability, and system-level constraints before entering preliminary design. Design options were compared using weighted trade matrices, which quantified performance across key FoM. The following section details the trade studies performed, the candidate configurations considered, and the rationale for the selected design.

FoM	Factor	Factor Justification
Integration	3	Integration quality determines whether subsystem performance is realized at the aircraft level.
Mission Performance	3	The performance of individual mission components directly affects TMS.
Aerodynamics	2	Drag reduction improves cruise and tow performance, but with diminishing returns compared to mission performance.
Stability	2	Adequate stability is required for repeatable missions but small changes within the desired ranges have minimal effects.
Weight	2	Increased weight affects stall speed and decreases payload capacity across missions, but promotes structural robustness.
Manufacturing	1	Primarily impacts iteration rate and schedule, not TMS directly.

Table 3.3.1: FoM justification

3.3.1 Aircraft Trade Studies

Aircraft Structure Five aircraft configurations were evaluated in **Table 3.3.2**: flying wing, lifting body, biplane, twin-boom, and single-boom. While the flying wing and lifting body offer reduced drag, both provide limited internal volume, reduced inherent stability [8], and control authority [9]. The biplane offers improved low-speed lift and structural strength, but incurs increased drag, mass, and integration complexity. A twin-boom configuration improves rotational stability, beneficial for banner deployment in M3, but introduces additional weight, drag, and reduced control efficiency due to HT-VT flow separation. The single-boom configuration was selected as it provides the most favorable balance of manufacturability and mission compatibility, while allowing for greater design flexibility to control weight, drag, and stability.

						
FoM	Factor	Flying Wing	Twin-boom	Single-boom	Lifting Body	Biplane
Mission Performance	3	1	4	5	1	3
Aerodynamics	2	5	3	4	4	2
Manufacturing	1	2	3	4	1	2
Total Score		15	21	27	12	15

Table 3.3.2: Aircraft structure matrix

Wing Location Configuration Three wing configurations were considered: high-wing, mid-wing, and low-wing. The high-wing configuration offers superior roll stability and provides increased fuselage storage capacity compared to other configurations. A low-wing could require a dihedral to offset lateral instabilities, which would increase manufacturing complexity. Additionally, a low-wing would interfere with landing gear mounted on the fuselage bottom. A mid-wing configuration would require a spar through the fuselage, disrupting payload storage. All configurations have similar manufacturing difficulty, but the high-wing offers the simplest integration with the fuselage while maximizing the fuselage's available internal volume. **Table 3.3.3** depicts this trade study.

				
FoM	Factor	High-wing	Mid-wing	Low-wing
Mission Performance	3	4	2	3
Integration	3	4	3	3
Aerodynamics	2	3	3	4
Stability	2	4	3	2
Total Score		38	27	30

Table 3.3.3: Wing configuration matrix

Tail Configuration Three tail configurations were considered: conventional tail, T-tail, and cruciform tail, as shown in **Table 3.3.4**. The scores were calculated based on a high-wing placement. The conventional tail was selected because it integrated most robustly, even though airflow over it is affected by the turbulent wake from the main wing. The T-tail's primary advantage is that it is outside of the downwash produced by wingtip vortices. However, the T-tail requires a stronger vertical structure to prevent bending, is heavier, and is more difficult to manufacture. The cruciform tail offers some of the same aerodynamic benefits as the T-tail, but still presents manufacturing and integration challenges. Given that integration was the primary focus, this reduced its overall desirability.

				
FoM	Factor	Conventional	T-tail	Cruciform
Integration	3	4	2	3
Aerodynamics	2	3	4	4
Manufacturing	1	4	2	3
Total Score		22	16	20

Table 3.3.4: Tail configuration matrix

Motor Three propulsion configurations were considered, detailed in **Table 3.3.5**: single tractor, twin pusher, and twin tractor. The twin tractor configuration offered increased lift through propeller slipstreams over the wing and reduced torque-induced roll and p-factor; however, these benefits were offset by increased mass, integration difficulty, and reduced propulsive efficiency. The single tractor configuration avoided these penalties, though it did not mitigate p-factor as effectively. The twin pusher configuration minimized propeller–fuselage interaction but did not offer aerodynamic advantages sufficient to justify its integration challenges. The single tractor configuration was selected, as it minimized system weight and integration complexity while operating in undisturbed airflow, which resulted in improved overall aerodynamic performance.

				
FoM	Factor	Single Tractor	Twin Pusher	Twin Tractor
Integration	3	4	2	2
Aerodynamics	2	4	3	3
Weight	2	4	2	2
Stability	2	3	3	4
Total Score		34	22	24

Table 3.3.5: Propulsion configuration matrix

Landing Gear Configuration Three landing gear configurations were considered: tricycle, taildragger, and quadricycle. The trade study is shown in **Table 3.3.6**. A tricycle configuration was chosen because it allows for maximum payload accessibility during GM, boosting its mission performance, while performing similarly in the aerodynamics categories to a taildragger. Although slightly less stable during takeoff and landing than taildragger and quadricycle configurations, prior competitions in Wichita demonstrate that a well executed tricycle is sufficiently stable.

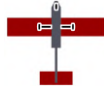
				
FoM	Factor	Tricycle	Taildragger	Quadricycle
Mission Performance	3	5	3	2
Aerodynamics	2	4	4	2
Stability	2	3	4	4
Manufacturability	1	4	3	2
Total Score		33	28	20

Table 3.3.6: Landing gear configuration matrix

3.3.2 Mission Trade Studies

Passenger Configuration Three restraint concepts were considered: a contoured foam block, a netted clamp, and a stretched clamp. The foam block provided secure, lightweight restraint but increased unloading time due to its size. A netted clamp caused passenger entanglement and inconsistent removal, while a continuous elastic clamping surface enabled the fastest and most repeatable loading and unloading. Based on the weighted trade study shown in **Table 3.3.7**, the stretched clamp configuration best supported the objective of minimizing GM time and ensuring passengers remain secure.

				
FoM	Factor	Foam Blocks	Netted Clamp	Stretched Clamp
Mission Performance	3	2	1	5
Weight	2	3	4	4
Manufacturability	1	4	3	3
Total Score		16	14	26

Table 3.3.7: Passenger configuration matrix

Banner configuration Three banner configurations were considered: an under-wing, under-tail, and hybrid configuration. The under-wing configuration showed the greatest support during banner deployment, but is hugely detrimental to aircraft stability. The hybrid configuration is moderately stable but is difficult to access, leading to a slower GM performance. The under-tail configuration was selected, as outlined in **Table 3.3.8**, for its minimal effect on aircraft stability when stowed and its accessibility during GM.

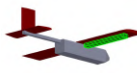
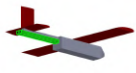
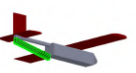
				
FoM	Factor	Under-wing	Under-tail	Hybrid
Mission Performance	3	3	4	3
Stability	2	1	4	3
Manufacturability	1	3	3	2
Total Score		14	23	17

Table 3.3.8: Banner configuration matrix

3.4 Design Concept

The design concept employs a high wing, single-boom conventional tail, tricycle landing gear configuration, and single tractor propeller to best satisfy flight requirements in **Table 3.3.1**. To optimize mission performance, the aircraft also features stretched-clamps to secure three passengers, a closed compartment to hold one cargo, and secures the **150.00 in** banner under the tail. The conceptual model for the aircraft is featured in **Figure 3.4.1**

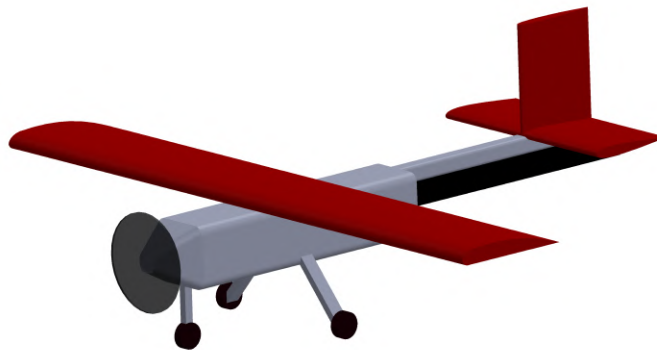


Figure 3.4.1: Selected design concept

4 Preliminary Design

4.1 Design and Analysis Methodology

CUDBF utilizes a systems engineering approach to design the competition aircraft. As shown in **Table 3.1.1**, the DBF 2026 rules were broken down into a series of system-level requirements. As determined by the sensitivity study discussed in **Section 3.2**, the five most sensitive design parameters were selected to be optimized by the mission model. GM time and wingspan had the most significant effect, followed by passenger count, cargo count, banner length, and battery capacity. The aspects of the aircraft determined by the trade studies in **Section 3.3** were fixed in the mission model. The constraints of the mission model's design space were determined based on the competition requirements and estimates made by the team, the methodology for which will be discussed in **Sections 4.2 and 4.3**.

The mission model is a MATLAB-based [1] high-fidelity multi-disciplinary design optimization framework that simulates the competition mission profiles and finds an optimal aircraft geometry with the five critical parameters acting as primary design variables. A model of the best-scoring feasible design was then constructed in XFLR5 [3], and a stability analysis was conducted. The results of this analysis were used to verify the aircraft's airworthiness and flight characteristics. Additionally, several propulsion system configurations were simulated using eCalc [4] to determine the most suitable one for the optimized design. If the produced design met all the requirements, a CAD model was created and fabricated to verify the validity of mass estimates and perform ground and flight tests. Testing data were used to validate performance characteristics and refine the mission model. Adjustments were made using the iterative process outlined in **Figure 4.1.1**.

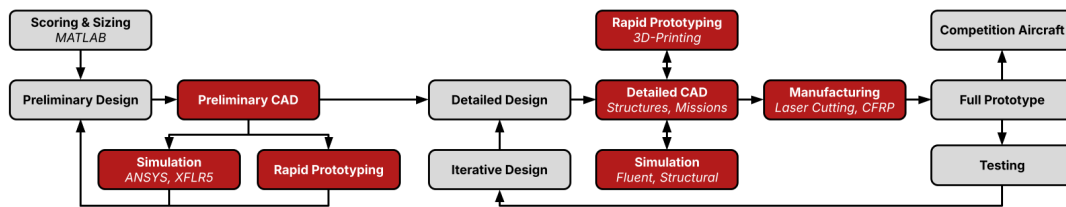


Figure 4.1.1: Design process and procedures

4.2 MDO Mission Model

The mission model employs a MDO framework coupling aerodynamic, structural, and propulsion analyses with external data to evaluate performance within competition constraints. The framework is implemented in MATLAB and organized into modular subsystems that exchange data and perform verification tasks. A 31-variable design vector, defining aircraft geometry and configuration, is passed through each analysis module and ultimately evaluated by a genetic algorithm to identify high-performing designs. The module structure and data flow are shown in **Figure 4.2.1**, with **Table 4.2.1** summarizing the function and objectives of each.

The outputs of the MDO framework directly informed the preliminary design of the competition aircraft, which is detailed in the subsequent sections. The mission course geometry used to simulate each competition run within the program is also shown in **Figure 4.1.2**.

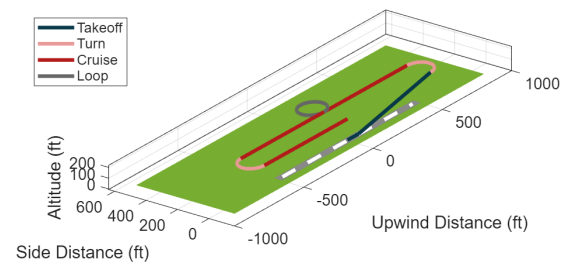


Figure 4.1.2: Mission model virtual course

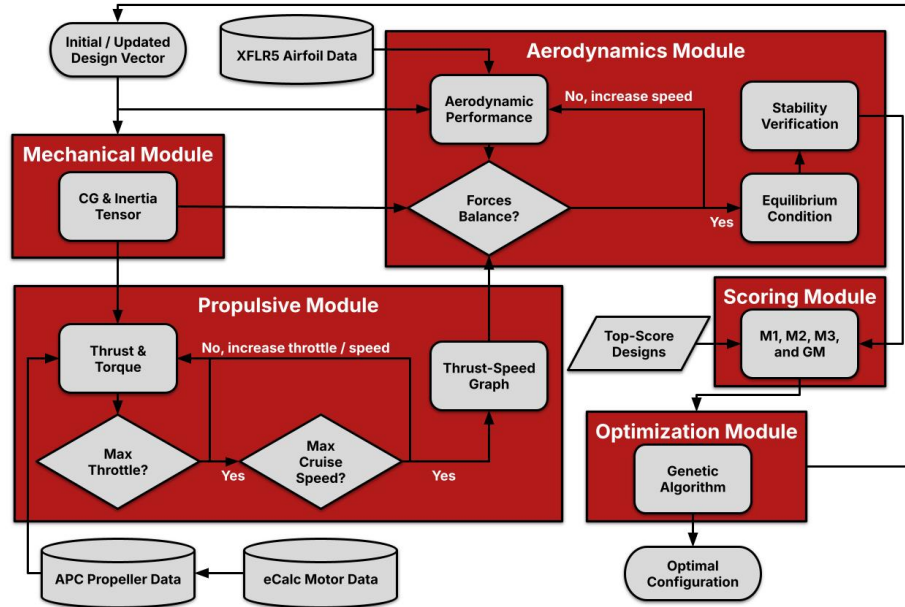


Figure 4.2.1: MDO model code diagram

4.3 Aircraft Sizing Procedure

To translate the results of the MDO mission optimization into a physically realizable aircraft configuration, the final population of evaluated designs was analyzed to identify dominant sizing trends and performance tradeoffs. PCA was performed on the optimization dataset to verify the parameter relationships identified in the sensitivity study. The analysis, illustrated in **Figure 4.2.2**, revealed five dominant mission factors consistent with the sensitivity results, while mechanical power emerged as an additional principal component being a proxy for cruise velocity. These six variables are used as the primary axes for the visualization shown below. **Figure 4.3.1** presents the genetic algorithm optimization results as a parallel coordinates plot, where each line represents a candidate design generated during the analysis. The lines intersect each axis at the corresponding variable value, and line color denotes TMS. Over 150,000 designs were evaluated, though only feasible configurations are shown for clarity.

Two primary design trends towards a high TMS emerged. The first favors M2, featuring a low-capacity battery, shorter banner, and greater payload. The second prioritizes GM and M3, employing a larger battery and longer banner while minimizing passengers and cargo. Consistent with the sensitivity analysis, reducing RAC yields a higher TMS and it is more effective to optimize for the second primary design trend. Additionally, the M3 cruise velocity trended near the stall speed for all designs. This pattern demonstrated that maximizing M3 score depends

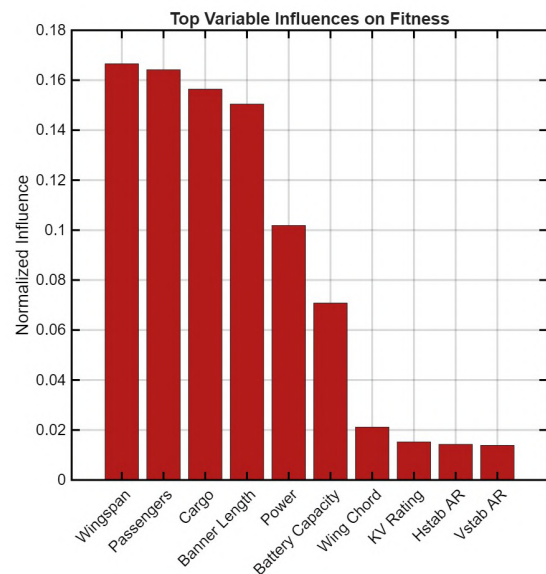


Figure 4.2.2: Mission model virtual course

Module	Objective
Aerodynamic	Model aerodynamic properties of aircraft wings, empennage, and fuselage. Combine geometric parameters with propulsion performance and mass properties, to compute lift, drag, and aerodynamic moments across the operating envelope. Verify static and dynamic stability by enforcing stall margin, thrust, and control constraints.
Mechanical	Estimate the mass given dimensions and configuration using empirical models. Calculate the aircraft's CG via a weighted average of individual components. Compute the moment of inertia tensor of the aircraft using the parallel axis theorem.
Propulsion	Given an incoming airspeed and throttle, calculate the propulsion system's thrust and power. Identify the characteristic thrust-velocity relationship. Evaluate aircraft flight time and battery endurance.
Scoring	Simulate performance across three flight missions and calculate TMS. Apply penalty functions to account for relative performance in negotiable categories, including dynamic flight behavior and usable flight envelope.
Optimization	Iteratively converge on optimal aircraft configuration using a genetic algorithm. Introduce sufficient variation in population to maximize design options considered.

Table 4.2.1: MDO module responsibility breakdown

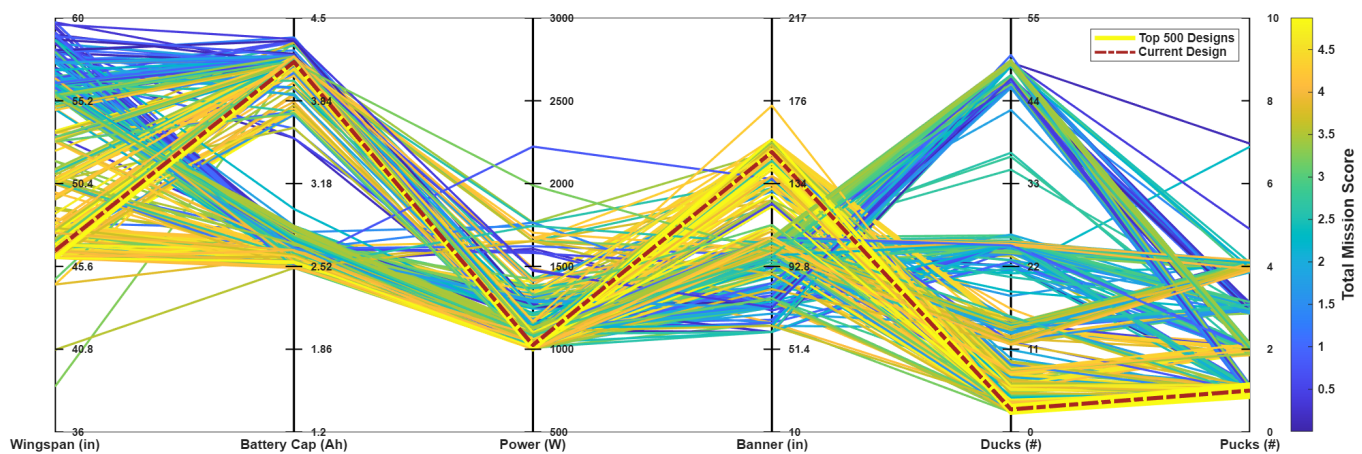


Figure 4.3.1: Desizing parameter scoring study

on maximizing banner length, reducing stall speed, and increasing available thrust. The sizing program ultimately identified an optimal configuration with **3 passengers**, **1 cargo**, and a **150.00 in** banner. These mission parameters drive the preliminary aircraft design, detailed in the rest of **Section 4.3**.

4.3.1 Wing Sizing

Wing sizing was performed within the MDO model, with wingspan as a design variable. For each candidate configuration, wing area was adjusted to satisfy **Equation 10** at the desired flight condition while remaining consistent with propulsion capability and aircraft weight estimates. Feasibility boundaries were established using flight performance derived from prior CUDBF flight data and historical competition reports, defining the acceptable performance envelope shown in **Figure 4.3.2**, alongside the previous year's competition aircraft for reference.

Once the required wing area was established, a wingspan was converged to considering the tradeoff between induced drag and RAC. At a fixed wing area and lift coefficient, induced drag is proportional to the inverse of wingspan,

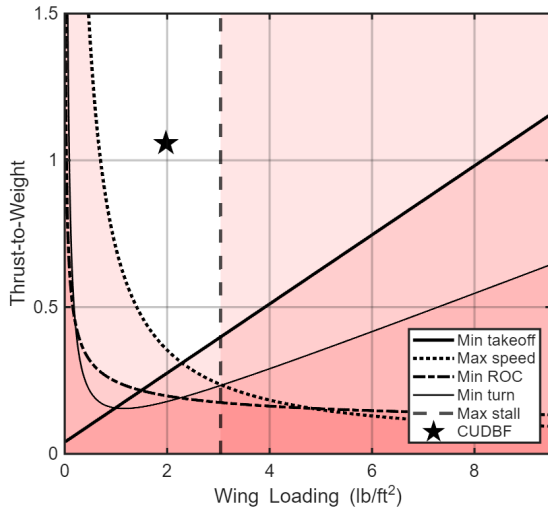


Figure 4.3.2: Constraint diagram

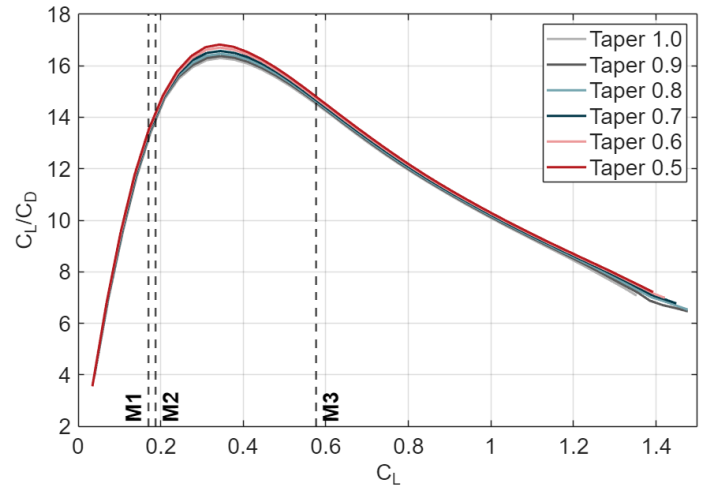


Figure 4.3.3: Wing taper graph

as seen in **Equation 11**. However, wingspan simultaneously increases RAC. Ultimately the MDO outputted an optimal wingspan of **46.51 in** and wing area of **562.31 in²**. The effect of wing taper on the aircraft's aerodynamic efficiency was investigated in XFLR5 for taper ratios from **0.5** to **1** at a fixed S_w . As shown in **Figure 4.3.3**, the maximum improvement in C_L/C_D was only **1.96%** in M3. Consequently, no taper was implemented.

$$L = \frac{1}{2} \rho v^2 S_w C_L \quad (10)$$

$$D_{\text{induced}} = \frac{1}{2} \rho v^2 S_w \left(C_L^2 \frac{S}{\pi b^2} \right) \quad (11)$$

4.3.2 Empennage Sizing

Empennage sizing was also performed within the MDO model using the tail volume coefficient relations in **Equation 12** with horizontal and vertical coefficients of **0.70** and **0.04**, respectively [10]. For each iteration, longitudinal and lateral stability were evaluated, with wing and tail lift and pitching moments used to locate the neutral point. Designs were required to maintain positive static margin, with penalties applied outside a preferred range of **5–25%** [10]. Pitch and yaw damping were also checked to ensure acceptable dynamic stability and controllability.

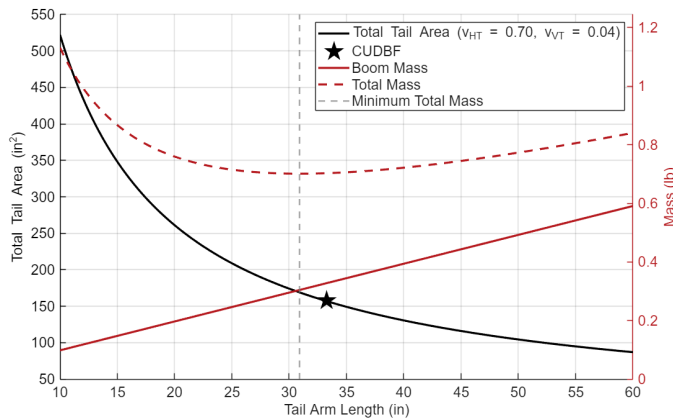


Figure 4.3.4: Empennage sizing tradeoffs

Tail arm length governs required tail area and corresponding tail mass, creating a tradeoff between empennage sizing and boom weight. Balancing this relationship, the empennage was positioned **33.27 in** aft of the wing leading edge. Incorporating stability constraints, the MDO model converged to a slightly longer tail arm than the purely mass-optimal solution, as shown in **Figure 4.3.4**. The resulting tail geometries employ a horizontal stabilizer span of **17.17 in** with an AR of **2.99**, and a vertical stabilizer height of **10.24 in** with an AR of **1.78**, values consistent with empirical design guidance from Raymer [10] and prior DBF aircraft. Using the same reasoning applied in **Section 4.3.1**, no taper was implemented in either of the stabilizers.

$$\bar{v}_{HT} = \frac{S_{HT} l_{HT}}{S_w \bar{c}_w} \quad \bar{v}_{VT} = \frac{S_{VT} l_{VT}}{S_w b_w} \quad (12)$$

4.3.3 Landing Gear

The location of the tricycle gears were determined based on methods described in Raymer [10]. This sizing led to the main landing gear supporting 85% of the aircraft's weight, with the front landing gear supporting the rest. The final geometry yields a tailstrike angle of **19.65 degrees** and tipback angle of **10.00 degrees**, satisfying ground-handling stability and rotation-clearance constraints. With the banner installed for M3, the tailstrike angle is reduced to **12.40 degrees**. These parameters are illustrated in **Figure 4.3.5** and discussed later in **Section 4.4.8**.

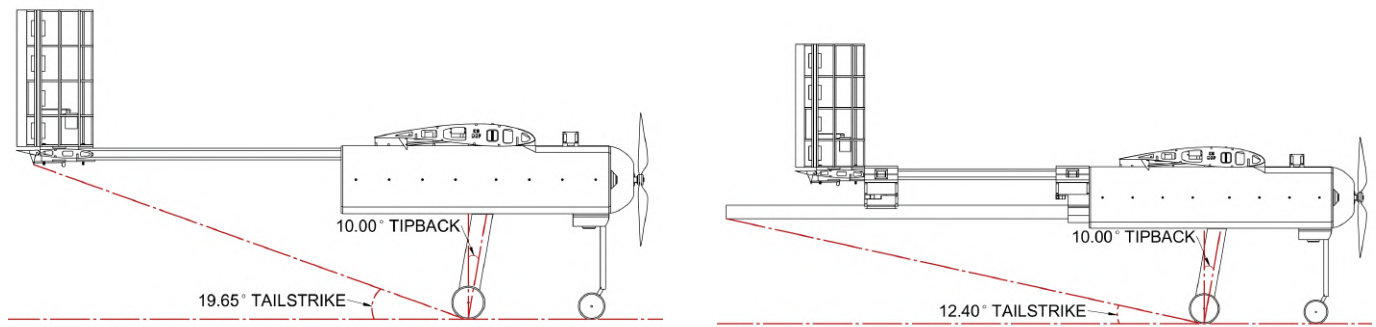


Figure 4.3.5: Landing gear geometry for M1 and M2 (left) and M3 (right)

4.3.4 Powerplant Sizing

The primary constraints on propulsion sizing were the M2 EF penalty and the **100 A** fuse limit, as defined in **Table 3.1.1**. Propulsion selection was embedded within the MDO model to identify motor–propeller–battery combinations capable of satisfying thrust requirements across the flight envelope. Battery voltage was fixed at **22.2 V** in the analysis to mitigate scoring penalties associated with stored energy, allowing battery capacity to emerge as an optimization output as shown in **Equation 13**.

$$P = C \times V \quad (13)$$

Figure 4.3.6 presents the propulsion constraint analysis as a function of flight speed. Lap count is normalized by the maximum achievable velocity of **140 ft/s**, while energy consumption and current draw are normalized by the competition limits of **100 Wh** and **100 A**. Using preliminary drag estimates, the study defines the trade between increasing lap potential and electrical limiting factors, establishing the feasible operational window. The previous year's CUDBF competition aircraft is included for reference.

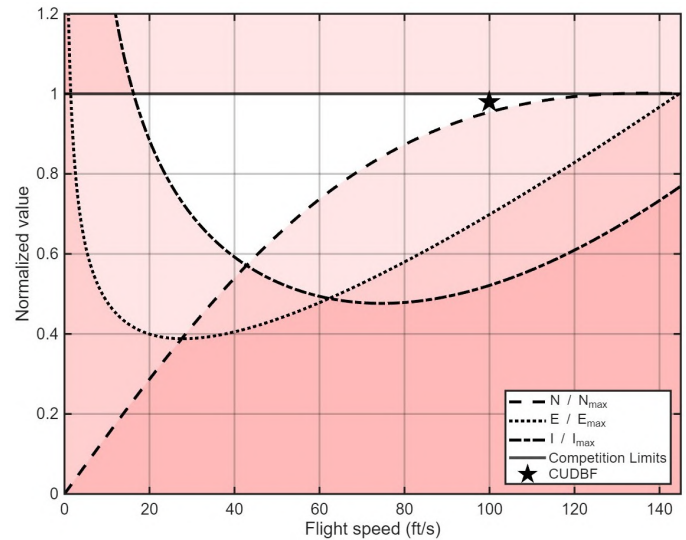


Figure 4.3.6: Propulsion constraints

The MDO converged to a narrow propulsion design space, defining viable ranges of motor Kv, power rating, propeller selection, and battery capacity independent of specific COTS components. These results established a baseline configuration, later refined through higher-fidelity analysis and integration constraints, and are summarized in **Table 4.3.1**.

4.3.5 Powerplant Selection

Because the MDO did not account for specific COTS components, a secondary powerplant selection process translated theoretical sizing outputs into commercially feasible options while preserving the baseline design space. Candidate motors, propellers, and batteries were selected based on proximity to MDO targets, with manufacturer datasheets providing motor constants, efficiency data, and operating limits. These parameters were implemented in MATLAB-based propulsion scripts [11] and cross-validated using eCalc propCalc [4] to predict thrust, shaft power, voltage, and current across the flight envelope. Motor speed and torque relationships were computed using **Equations 14** and **15**, with constants obtained from specific motors.

Parameter	Value
Diameter (in)	14.4
Pitch (in)	6.8
Power (W)	1050
Kv Rating	490
Capacity (mAh)	4100

Table 4.3.1: MDO ideal output

$$\omega_m = K_V \times V \quad (14)$$

$$Q_m = K_T \times I \quad (15)$$

$$t_{\text{flight}} = \frac{60}{1000} \times \frac{C}{I} \quad (16)$$

Predicted current draw was evaluated against the **100 A** fuse constraint, and achievable endurance was estimated using **Equation 16** to ensure compliance with both mission duration and energy limitations. This process narrowed the propulsion options to a small set of commercially available systems that closely matched the MDO ideal while remaining feasible from integration and availability perspectives. The selected configuration shown in **Table 4.3.2** established the baseline for testing described in **Section 7** and was carried forward into detailed design.

Propulsion Component	Specifications	
	M1 & M3	M2
Propeller	15 in x 6 in	14 in x 7 in
Motor	Scorpion SII-4025-520KV 1200W	
Battery	Tattu 4200mAh 6s 35c	Lipierior 4000mAh 6s 30c

Table 4.3.2: Preliminary propulsion configuration

4.3.6 Banner Parameters

Banner Sizing The drag generated by a banner flying through the air is approximated by **Equations 17** and **18** using an estimated M3 cruise speed of **52.5 ft/s**.

$$D = \frac{1}{2} \rho v^2 S_{\text{wet}} C_f \quad (17)$$

$$C_f = \frac{0.455}{(\log_{10} Re)^{2.58}} - \frac{3143}{Re} \quad (18)$$

The drag generated versus banner length for different aspect ratios is shown in **Figure 4.3.7**. To account for differences between the idealized flat plate model described by **Equation 18** and the fluttering actual flight of the banner, a corrective factor of **20** is applied to the predicted drag determined by **Equation 17** before plotting and during further analysis. This factor was determined through the small scale wind tunnel testing described in **Section 8.1.6**.

A higher AR banner reduces aerodynamic drag, so the maximum allowable AR of **5.00** is used. The MDO model converged to **150.00 in** as the optimal banner length. **Equation 19** illustrates the power dissipated by the banner in flight, which at **1000 W** aligns with predicted propeller power output of **1050 W**.

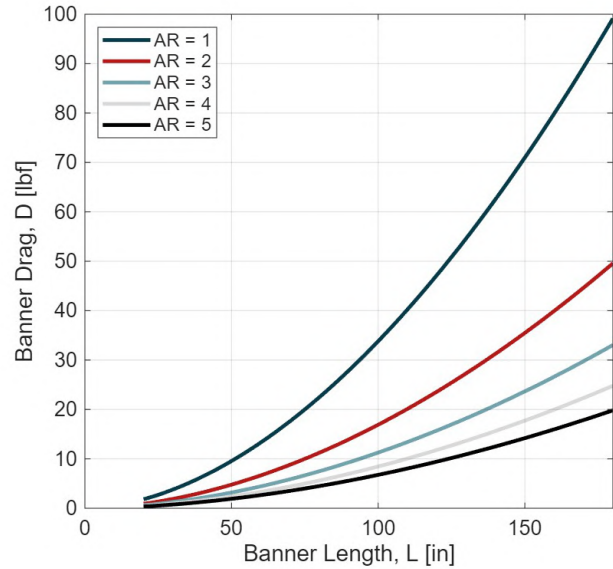


Figure 4.3.7: Banner drag vs length at various AR

$$P_{dissipated} = D \times v \quad (19)$$

Material Choice Material selection prioritized low mass and streamlined aerodynamics, leading to the selection of ripstop nylon from other materials evaluated in **Table 4.3.3**. Its thickness minimized oscillatory behavior in flight [7] and the material still maintained a low surface roughness, which decreased its skin friction drag. Further material testing to refine drag expectations is detailed in **Section 8.1.6**.

Material	Ripstop Nylon	LDPE	Vinyl PVC Sheet
Thickness	0.002 in	0.004 in	0.02 in
Weight per square foot	0.00983 lb	0.0278 lb	0.10 lb
Surface Roughness	787.40 μ in	492.13 μ in	118.11 μ in

Table 4.3.3: Banner potential material properties

4.4 Aircraft Analysis Procedure

4.4.1 Airfoil Selection

The wing airfoil was selected using a weighted trade matrix to quantitatively compare candidates based on aerodynamic performance and practical considerations. $C_{L,max}$ and manufacturability were weighted the highest because $C_{L,max}$ directly governs stall speed, which is critical when flying with the banner drag in M3, while manufacturability strongly influences achievable surface quality and integration with internal components such as the servos and spar. C_L/C_D was assigned a moderate weight since the overall aircraft configuration also influences this ratio. C_{L0} and stall angle were weighted the lowest because they were similar across all candidates and are less critical given the lack of a takeoff distance limit.

Four candidate airfoils were selected from a larger pool: FX 60-126, MH-114, NACA 2412, and USA-35B. These airfoils were analyzed at the expected Reynolds number of **300,000** using XFLR5 as shown in **Figure 4.4.1**, with the comparative assessment identifying the FX 60-126 as a strong candidate.

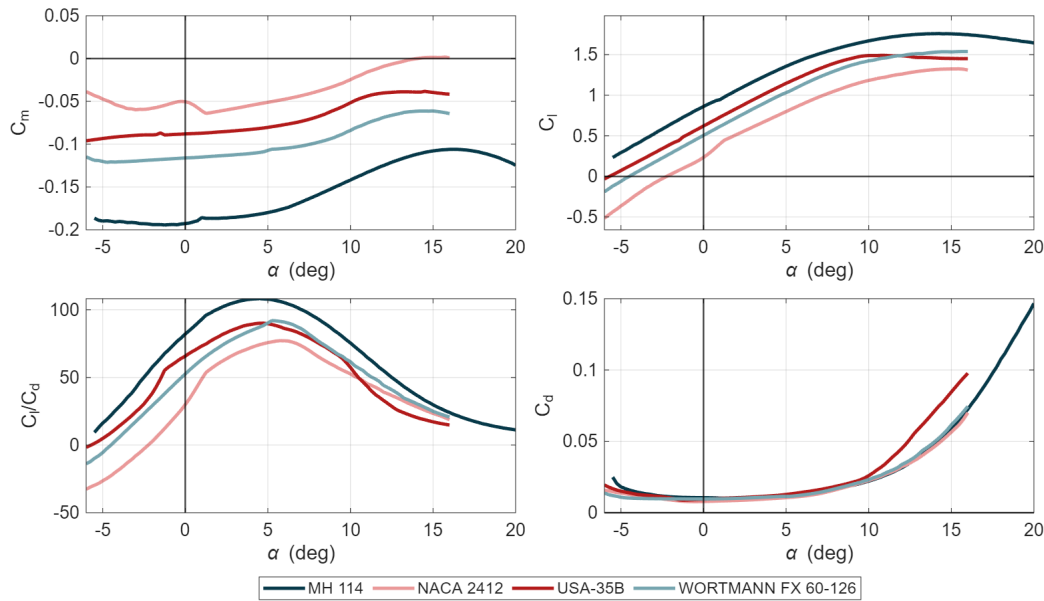


Figure 4.4.1: Aerodynamic performance curves for considered airfoils

					
FoM	Factor	FX 60-126	MH-114	NACA 2412	USA-35B
$C_{L,max}$	3	4	5	1	2
Manufacturability	3	4	2	4	5
$C_L/C_{D,max}$	2	4	5	2	4
C_{L0}	1	3	4	2	3
Stall angle	1	5	4	4	2
Total Score		40	39	25	34

Table 4.4.1: Airfoil decision matrix

4.4.2 Static Stability and Flight Characteristics

XFLR5 was used to evaluate static stability and flight characteristics for each mission. Mission-specific mass distributions were derived from a preliminary CAD model, incorporating components and mission-dependent payloads into the simplified point-mass representation shown in **Figure 4.4.2**.

The absolute forward and aft CG limits are shown in **Figure 4.4.3**. The CG must be forward of the neutral point to achieve longitudinal static stability, so the neutral point is set as the aft limit. The forward limit considers the need for sufficient control authority, and was calculated using **Equation 20** to be **3.90%** of the wing chord in front of the wing leading edge.

$$x_{CG,min} = x_{NP} - \frac{C_{m_0} + C_{m_{\delta e}} \delta e_{min}}{C_{L_{max}} - C_{L_0} - C_{L_{\delta e}} \delta e_{min}} \quad (20)$$

While these represent the absolute CG limits, a SM range of **5-25%** is desirable to balance stability and controllability, based on RC aircraft convention [12] and previous team experience. XFLR5 was used to validate the SM of

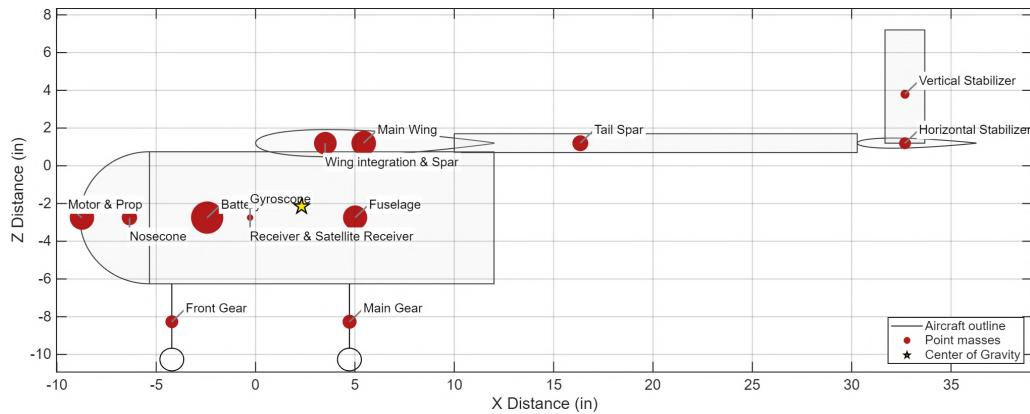


Figure 4.4.2: M1 point-mass distribution diagram

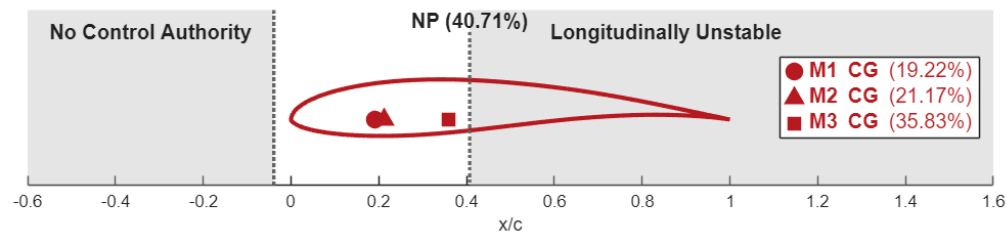


Figure 4.4.3: DF1 absolute CG limits

the design generated by the MDO. Though M3 displayed a significant drop in SM, all mission configurations had SMs within the desirable range. A **2 degree** downward angle of incidence was implemented on the horizontal stabilizer to ensure that an equilibrium condition could be maintained in flight. This angle was determined by an iterative analysis process in XFLR5.

The **Table 4.4.2** displays the cruise parameters for each mission, along with the static stability derivatives determined through XFLR5, which were deemed to be within an acceptable range as discussed by Sadraey [12].

Lastly, the aerodynamic performance curves for the three missions are shown in **Figure 4.4.4** below, with the cruise condition points marked with circles.

Mission	Mass Properties				Stability Derivatives			Velocities	
	α ($^{\circ}$)	NP (in)	Mass (lbm)	SM (%)	$C_{m\alpha}$ (l°)	$C_{l\beta}$ (l°)	$C_{n\beta}$ (l°)	V_{cruise} (ft/s)	V_{stall} (ft/s)
M1	-1.08	4.92	6.554	21.8	-1.48e-2	-1.08e-4	1.04e-3	90.88	31.69
M2	-0.82	4.96	7.228	20.0	-1.36e-2	-1.05e-4	1.04e-3	90.88	33.28
M3	4.96	5.51	7.366	9.7	-6.55e-3	3.00e-7	1.01e-3	52.48	33.60

Table 4.4.2: Cruise conditions and stability derivatives for each mission

4.4.3 Dynamic Stability Characteristics

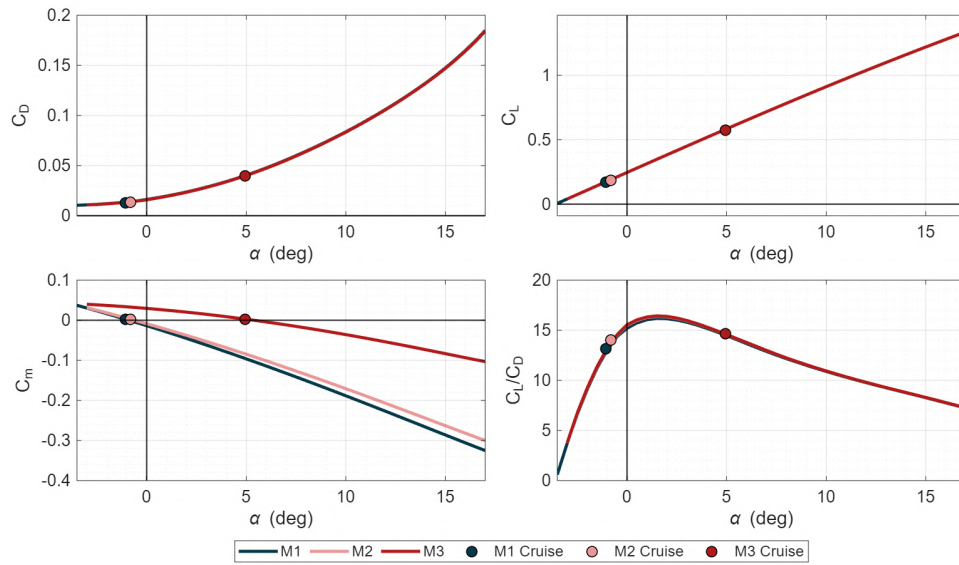


Figure 4.4.4: Flight missions aerodynamic performance

Mission	Parameter	Longitudinal Modes		Lateral Modes		
		Short Period	Phugoid	Roll Subsidence	Dutch Roll	Spiral Divergence
M1	Natural Frequency (Hz)	2.340	0.066	-	1.340	-
	Time to half/double/damping ratio	$\zeta = 0.597$	$\zeta = 0.010$	$t_{1/2} \approx 0.007$ s	$\zeta = 0.151$	$t_2 = 12.380$ s
M2	Natural Frequency (Hz)	2.231	0.066	-	1.329	-
	Time to half/double/damping ratio	$\zeta = 0.594$	$\zeta = 0.010$	$t_{1/2} = 0.007$ s	$\zeta = 0.149$	$t_2 = 11.610$ s
M3	Natural Frequency (Hz)	0.777	0.102	-	0.636	-
	Time to half/double/damping ratio	$\zeta = 0.688$	$\zeta = 0.005$	$t_{1/2} = 0.012$ s	$\zeta = 0.192$	$t_2 = 2.770$ s

Table 4.4.3: Dynamic stability characteristics of DF1

A dynamic stability analysis was conducted in XFLR5 for the flight conditions corresponding to all three missions. The aircraft equations of motion were linearized about steady flight, allowing disturbance response to be decomposed into standard longitudinal and lateral modes. Short-period and phugoid modes were used to assess longitudinal stability, while roll subsidence, Dutch roll, and spiral modes characterized lateral behavior. For acceptable dynamic stability, all modes must exhibit stable or controllable behavior.

Results of the analysis are summarized in **Figure 4.4.5** and **Table 4.4.3**. The aircraft is dynamically stable in the short-period, phugoid, roll subsidence, and Dutch roll modes. An unstable spiral mode was identified, consistent with the near-neutral roll stability; however, spiral divergence is common in payload-optimized RC-scale aircraft and is generally correctable through pilot input.

Considering both static and dynamic stability, M3 represents the most critical flight condition due to its higher mass, lower flight speed, higher angle of attack, and reduced spiral divergence time.

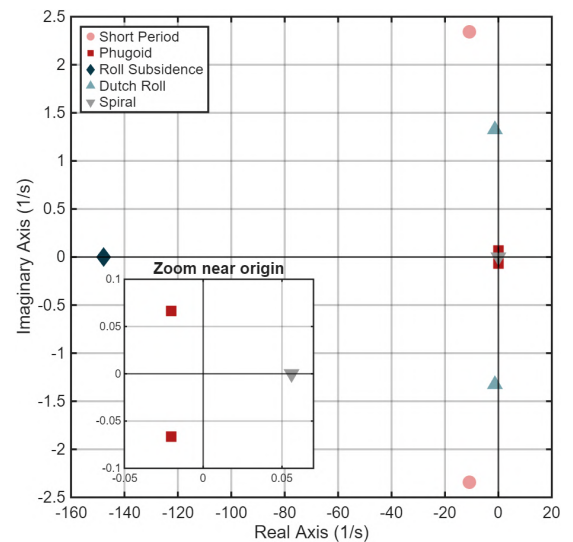


Figure 4.4.5: M1 aircraft dynamic mode poles

4.4.4 Controllability

Three control surfaces were implemented on the aircraft: ailerons, elevator, and rudder. Initial sizing was performed using methods described in Raymer [10] and geometric trends from historical DBF aircrafts. The elevator and rudder were sized with a **100%** span ratio and **30%** chord ratio, while the ailerons were sized with a **40%** span ratio and **25%** chord ratio. The maximum deflection for all control surfaces was limited to ± 30 degrees based on mechanical constraints of the actuators and hinge geometry.

Parameter	Variable	XFLR5 (l°)
Aileron Rolling Power	$C_{l\delta_a}$	-0.0040
Lift Variation with Elevator	$C_{L\delta_e}$	0.0072
Elevator Pitching Power	$C_{m\delta_e}$	-0.0206
Force Variation with Rudder	$C_{Y\delta_r}$	0.0011
Rudder Yawing Power	$C_{n\delta_r}$	-0.0008

Table 4.4.4: Key control derivatives for M1 cruise

XFLR5 was used to validate this sizing and assess the aircraft's controllability. The control derivatives were estimated from the moment coefficient curves at varying degrees of control surface deflections, with the results summarized in **Table 4.4.4**. Comparison with prior DBF aircraft indicates that the selected control surface dimensions provide adequate control authority and handling qualities across the expected operating envelope.

4.4.5 Turning Performance

Equations 21 and **22** were used to evaluate the turning performance of the aircraft, as illustrated in **Figure 4.4.6**. The figure presents turn rate as a function of airspeed, with the two families of curves corresponding to constant load factor and constant turn radius. As airspeed increases, structural considerations dominate, and the maximum allowable load factor governs the achievable turn rate. Based on RC aircraft convention and prior team experience, a standard banked turn was determined to be one with a maximum load factor of **2**. Constant-radius curves are included to provide geometric context for the resulting maneuvering envelope.

$$R = \frac{v^2}{g\sqrt{n^2 - 1}} \quad (21)$$

$$\omega = \frac{g\sqrt{n^2 - 1}}{v} \quad (22)$$

4.4.6 Climb Performance

ROC was calculated for each mission using **Equation 23**. **Figure 4.4.7** contains the results of these calculations. The ROC for each mission fell within the expected range based on previous DBF competition aircraft and team research.

$$\text{ROC} = \frac{v(T - D)}{W} \quad (23)$$

4.4.7 Aircraft Aerodynamics

A comprehensive drag characterization of DF1 was performed using a combination of empirical analytical methods and numerical simulations. Total drag was decomposed into induced drag and parasitic drag and the overall drag coefficient was expressed as shown in **Equation 24**.

$$C_D = C_{D_0} + \frac{C_L^2}{\pi e AR} \quad (24)$$

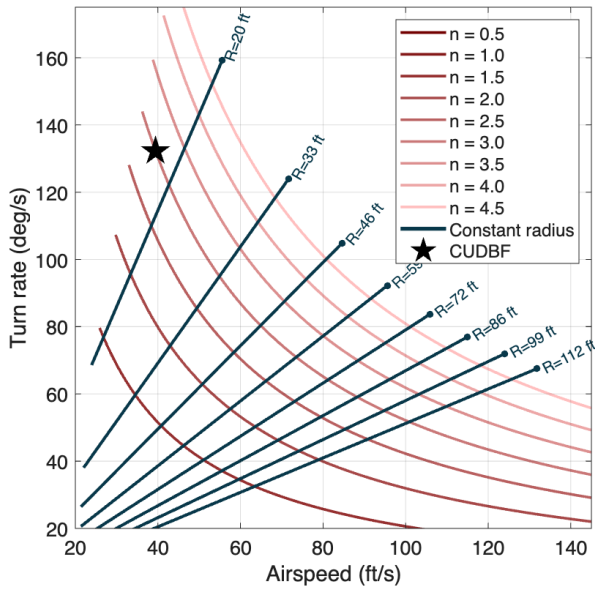


Figure 4.4.6: DF1 turn graph

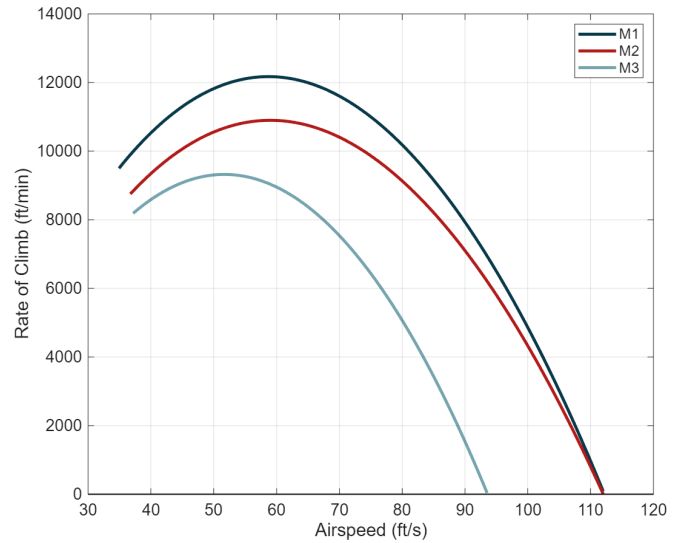


Figure 4.4.7: Rate of climb vs. velocity

An Oswald efficiency factor of **0.75** was selected for DF1's low-aspect-ratio rectangular wing. Two independent methods were used to determine and cross-validate the parasitic drag coefficient. First, OpenVSP [13] was used to generate component-level parasitic drag estimates for each mission configuration, shown **Figure 4.4.8**.

Second, a full-aircraft CFD analysis was conducted in ANSYS Fluent [5] using a steady Reynolds-averaged Navier–Stokes approach with a $k - \omega$ turbulence model. This analysis served both to corroborate the C_{D_0} predicted by OpenVSP and to characterize the wake structure relevant to banner placement, discussed further in **Section 5.2.1**. Simulations were performed at several low angles of attack, and the resulting lift and drag coefficients were extracted to construct a low-lift drag polar using a parabolic fit, yielding a C_{D_0} of **0.059**. This closely matched the OpenVSP estimate, providing confidence in the drag model used for subsequent mission analyses. Using this model and XFLR5 induced drag information, mission-specific drag polars were generated in **Figure 4.4.9**. For M3, additional drag from the banner was incorporated using the method described in **Section 4.3.6**. Due to the similarity of aircraft configurations, the M1 and M2 drag polars were found to be effectively identical.

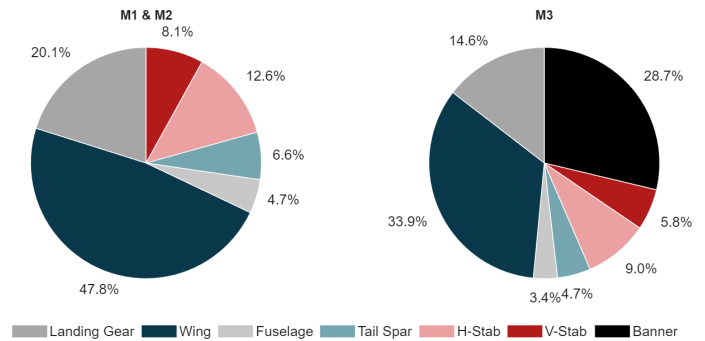


Figure 4.4.8: Component mission drag breakdowns

4.4.8 Takeoff Performance

With an estimated runway length of **300 ft**, the takeoff condition is primarily constrained by the requirement that the tail or banner must not strike the ground. Based on the preliminary aircraft geometry, the strike angle is **19.65 degrees** for M1 and M2, and **12.40 degrees** for M3.

Since the stall angle is **17.00 degrees**, as derived from XFLR5, takeoff for M1 and M2 is stall-limited rather than tail-strike limited. The takeoff velocity was thus estimated using the convention shown in **Equation 25**. **Equation**

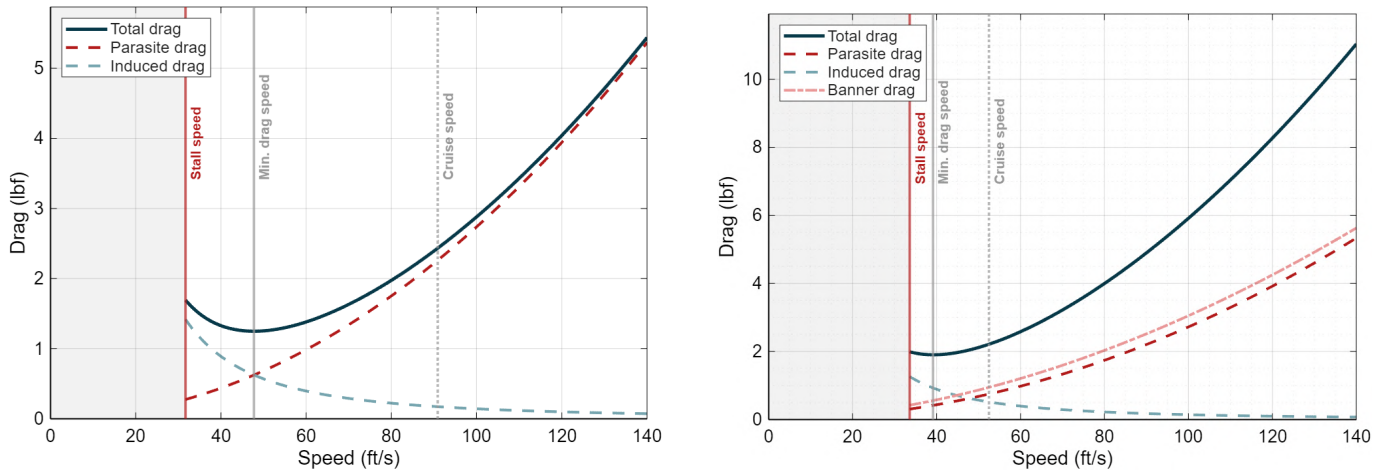


Figure 4.4.9: Drag polars for M1 and M2 (left) and M3 (right)

26 is used to determine the takeoff C_L , specifying alpha. For M3 the strike angle is used as the takeoff AoA, from which the takeoff C_L and thus the takeoff velocity are derived. The takeoff distance can be found using the average acceleration method as shown in **Equation 27**. **Equation 28** was used to calculate the elevator deflection required to achieve pitching moment equilibrium at these limiting AoA. The results of these calculations are summarized in **Table 4.4.5**, confirming that the aircraft can achieve the required takeoff velocities and elevator deflections.

$$v_{\text{takeoff}} = 1.2v_{\text{stall}} \quad (25)$$

$$C_L = \frac{2mg}{\rho v^2 S_w} \quad (26)$$

$$a = \frac{T - D - \mu(W - L)}{m} \quad (27)$$

$$C_m(\alpha_{\text{takeoff}}) + C_{m_{\delta_e}} \delta_e = 0 \quad (28)$$

Mission	Takeoff α (°)	Velocity (ft/s)	Distance (ft)	δ_e (°)
M1	11.02	38.02	22.40	11.55
M2	11.02	39.94	24.78	10.63
M3	12.40	39.24	23.89	4.27

Table 4.4.5: Takeoff conditions for each mission

A crosswind condition was also analyzed for takeoff and landing, assuming the aircraft remains aligned with the runway. The simplified yaw moment balance in **Equation 29** as guided by Sadraey [12] was used to determine the required rudder deflection for crosswind operation, including a corrective factor to account for propwash. At the maximum rudder deflection of **30 degrees**, the aircraft can safely operate in crosswinds up to **29.46 ft/s**.

$$\frac{1}{2}\rho v^2 S_w (C_{n_\beta} \beta + C_{n_{\delta_r}} \delta_r) + \frac{1}{2}\rho v_{\text{wind}}^2 S_{\text{side}} C_{D_y} (\bar{x} - x_{\text{CG}}) = 0 \quad (29)$$

4.4.9 Uncertainties

Uncertainty is inherent in preliminary analyses, and overly optimistic assumptions were avoided. Many of the analytical relations used in this section rely on simplifying assumptions, such as small-angle approximations, which limit their validity to specific operating regimes. Similarly, XFLR5 employs a vortex lattice method that does not accurately capture post-stall behavior. The CFD analyses conducted in ANSYS Fluent also incorporate modeling simplifications associated with steady Reynolds-averaged Navier–Stokes formulations and the $k-\omega$ turbulence model, and therefore do not capture unsteady flow phenomena such as vortex shedding that may influence lift and drag. These limitations were considered when interpreting results and informed the conservative design margins applied throughout the aircraft development process.

4.4.10 Risk Analysis

The operation of DF1 poses many risks, which are summarized and visualized in **Table 4.4.10** and **Figure 4.4.11**, respectively. The table indicates the risk index of each potential issue as the product of likelihood and severity. The table also presents a control plan for each risk.

Risk	Risk Index	Control Plan	Code
Loss of radio connection	$2 \times 5 = 10$	Extra satellite receiver	R1
Landing gear failure	$2 \times 4 = 8$	Reinforcement and testing	R2
Propulsion system failure	$2 \times 4 = 8$	Use historically proven products	R3
Banner release fault	$3 \times 2 = 6$	Extensive real and simulated testing	R4
Fuselage structural failure	$1 \times 5 = 5$	FEA and structural testing	R5
Stall during banner tow	$1 \times 4 = 4$	Cruise and stall speed buffer	R6

Figure 4.4.10: Risk analysis mitigation

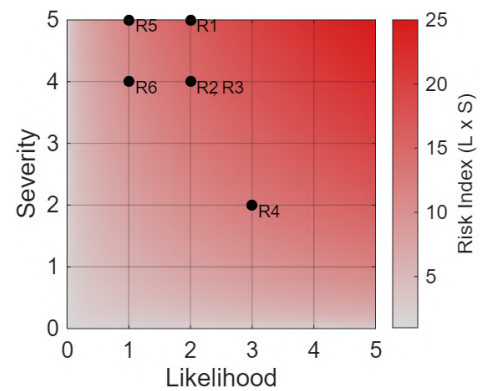


Figure 4.4.11: Risk index plot

4.5 Estimated Mission Performance

Based on the results of the trade studies and the preliminary design of the aircraft, the expected performance of the aircraft was calculated based on the mission scoring equations outlined in **Section 3.1.3**. These predicted mission scores are shown in **Table 4.5.1**.

Mission	Expected Score
M1	1.00
M2	1.02
M3	2.30
GM	0.58

Table 4.5.1: Estimated mission performance

5 Detailed Design

5.1 Aircraft Subsystem Design

5.1.1 Dimensional Parameters

Table 5.1.1 shows the final dimensions and characteristics for DF1.

Duck Force One					
Wing		Vertical Stabilizer		Horizontal Stabilizer	
Airfoil	FX 60-126	Airfoil	NACA 0012	Airfoil	NACA 0012
Span (in)	46.51	Span (in)	10.24	Span (in)	17.17
Chord (in)	12.09	Chord (in)	5.75	Chord (in)	5.75
AR	3.85	AR	1.78	AR	2.99
Planform Area (in ²)	562.31	Planform Area (in ²)	58.88	Planform Area (in ²)	98.73
Control Surface Ratio (%)	25%	Control Surface Ratio (%)	30%	Control Surface Ratio (%)	30%
Fuselage		Electronics		Incidence Angle (°)	-2.00
Total Length (in)	28.00	M1 & M3 Battery	Tattu 4200mAh 6s 35c LiPo	Propeller	
Height (in)	5.10	M2 Battery	Liperior 4000mAh 6s 30c LiPo	M1 & M3	15 in × 6 in
Width (in)	4.10	Receiver	Spektrum AR8020T	M3	14 in × 7 in
Motor		Servos	Xpert CI-2401	Banner	
Motor	Scorpion SII-4025-520KV	Nosecone		Length (in)	150.00
ESC	SunnySky X series 80A	Length (in)	1.95	AR	5.00

Table 5.1.1: Dimensional parameters

5.1.2 Operational limits

Across all three missions, the aircraft is subjected to hard landings, rapid turns, transient accelerations, and varying payload masses, resulting in multiple critical loading scenarios. The selected single-boom tricycle landing gear configuration informed an analysis of the resulting load paths through the airframe. These load paths, shown in **Figure 5.1.1**, were used to identify structurally critical regions and directly guided local reinforcement and structural sizing during the design process.

Structural design limits for DF1 were established using preliminary design flight predictions and maneuver load factors generated by the MDO model. A single aircraft-level V–n envelope was developed as shown in **Figure 5.1.2**, capturing both maneuver and gust-induced aerodynamic loading across the opera-

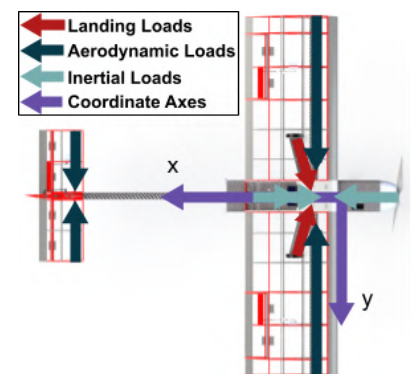


Figure 5.1.1: Load path diagram

tional speed range. A structural failure speed of **145 ft/s** was selected based on prior CUDBF experience. Additionally, a limit load factor of **3.5** was identified as the threshold at which subsystem joints and interfaces failed, despite individual components remaining structurally intact.

5.1.3 Fuselage Detailed Design

The fuselage was designed to minimize mass while maximizing usable internal volume without compromising structural integrity. The final configuration employs a CFRP monocoque, consisting of two 3-ply shells fastened together with nylon hardware. Each shell uses alternating **45 degree** ply orientations to improve torsional stiffness and load distribution.

The fuselage geometry consists of a **28 in** rectangular body with a **4.1 in × 5.1 in** cross-section and rounded edges to reduce aerodynamic drag. A **5.5 in × 3.5 in** upper avionics hatch and a **2.1 in × 3.6 in** inside cargo hatch provide access to internal systems; both are hinged and secured using snap fasteners for rapid operation.

Three basswood bulkheads subdivide the fuselage into electrical, cargo, and passenger compartments as shown in **Figure 5.1.3**, while providing mounting interfaces for internal systems and the nosecone. Basswood plates are used at the wing and tail interfaces to distribute concentrated loads into the CFRP shell.

Structural performance was verified using ANSYS ACP [5], evaluating a governing combined loading case consisting of a distributed wing attachment load, longitudinal electronics and tail-induced moments, and a landing load from the landing gear. The wing attachment load represented an extreme worst case scenario of **200 lbf**. All other loads were calculated based on the aircraft's structural limit load factor of **3.5**.

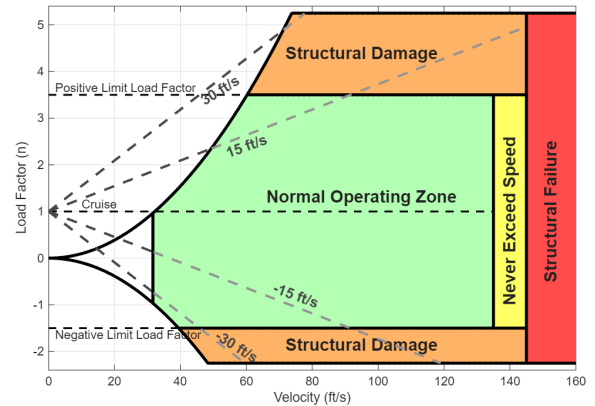


Figure 5.1.2: Structural flight envelope for DF1

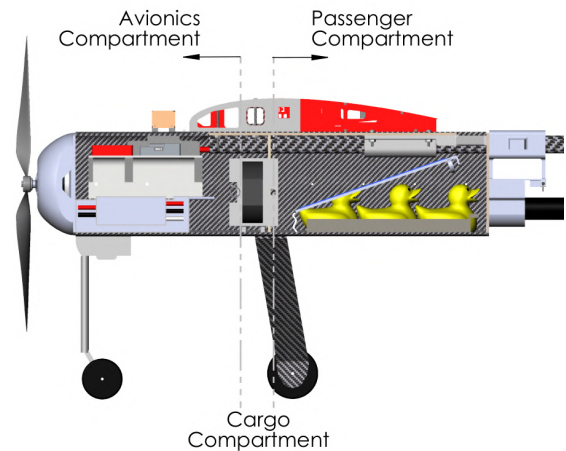


Figure 5.1.3: Monocoque airframe crosssection

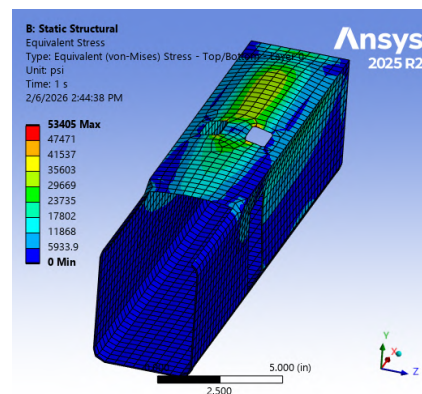
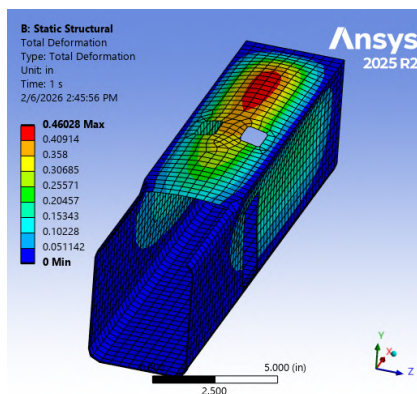


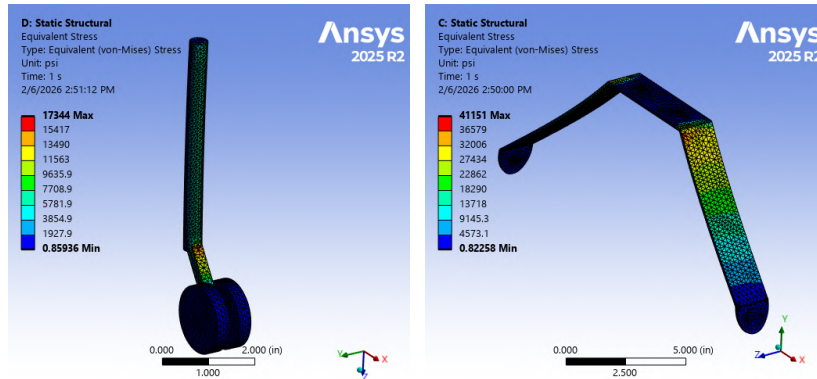
Figure 5.1.4: Deformation (left) and equivalent stress (right) in the fuselage

Preliminary results were evaluated using the Ansys Composite Failure tool and further studied using the equivalent stress. The composite failure tool indicated a very high FoS in the combined loading configuration. Additionally, evaluating the max stress against the **300,000 psi** ultimate strength of CFRP gives a specific FoS of **5.6** against this worst case loading scenario.

5.1.4 Landing Gear Detailed Design

The landing gear system consists of a carbon-fiber main gear frame to minimize mass while maintaining structural strength, and a steerable nose gear fabricated from a **0.40 in** diameter aluminum tube with spring-suspended wheels. Steering authority is provided by a dedicated servo housed within the nose gear assembly. Lightweight **2.5 in** foam wheels are used for all gear locations to reduce mass and rolling resistance.

The landing gear was sized to withstand landing loads corresponding to the MTOW of **7.4 lb**, using an assumed landing load factor of **3.5** as defined in **Section 5.1.2**. FEA was performed to verify that both the main and nose gear assemblies meet strength requirements, with representative stress distributions shown in **Figure 5.1.5**. The resulting structural margins are summarized in **Table 5.1.2**.



FoM	Main Gear	Front Gear
Load (lbf)	22.02	2.89
Max Stress (psi)	41,000	17,300
FoS	7.5	2.5

Table 5.1.2: Landing gear FEA results

Figure 5.1.5: Main gear (left) and front gear (right) stress.

The addition of landing gear fairings was evaluated as a potential aerodynamic improvement. As shown in **Figure 5.1.6**, a CFD analysis in ANSYS Fluent was conducted to quantify performance effects. The results indicated a minor drag reduction, with main gear drag decreasing from **0.38 lbf** to **0.37 lbf** (2.6%). A more substantial change was observed in lift, which increased from **0.08 lbf** to **0.27 lbf** (238%). Despite these aerodynamic gains, preliminary fairing designs observed a mass equivalent to **0.44 lbf** of additional lift requirement, offsetting the observed benefits. As a result, landing gear fairings were not incorporated into the competition aircraft.

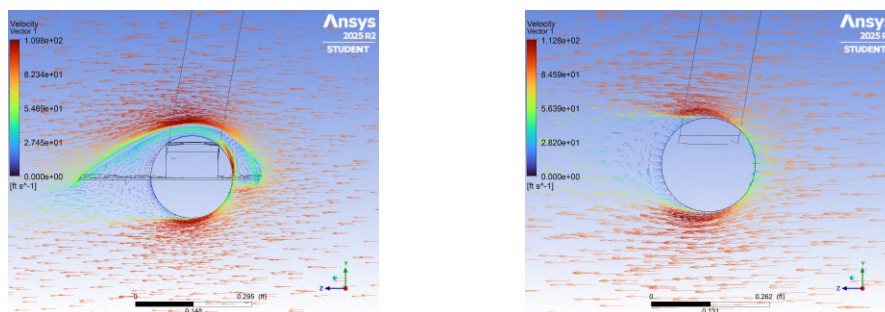


Figure 5.1.6: Landing gear CFD with (left) and without (right) fairings

5.1.5 Wing Detailed Design

The aircraft employs a rectangular planform high-wing configuration with no dihedral, no taper, and aileron control surfaces. Since there is no takeoff requirement, flaps or flaperons are not implemented. The wing and ailerons are designed as rib and stringer structures with one servo in each wing half. There is one square carbon fiber spar running through both wings to provide structural support. CA hinges are used to minimize control surface gaps, with balsa reinforcement blocks added to increase hinge bond area. To further minimize the control surface gaps, 3D printed PLA leading edges are implemented on the ailerons. An asymmetric triangular balsa piece forms the trailing edge, providing structural integrity while still conforming to the airfoil shape. The CAD design for the entire wooden wing structure and the servo mechanism was modeled in SOLIDWORKS [2], as shown in **Figure 5.1.7**.

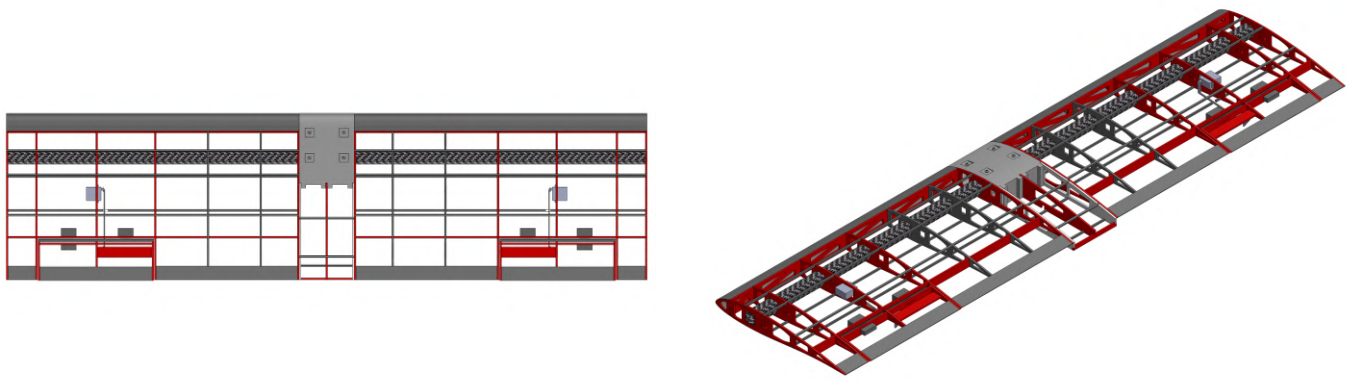


Figure 5.1.7: Wing CAD top view (left) and isometric view (right)

The potential use of endplates was evaluated through CFD analysis of multiple different geometries in ANSYS Fluent, with the results presented in **Figure 5.1.8**. The final design does not incorporate endplates, as the marginal aerodynamic benefits did not justify the added weight, manufacturing complexity, and build time.

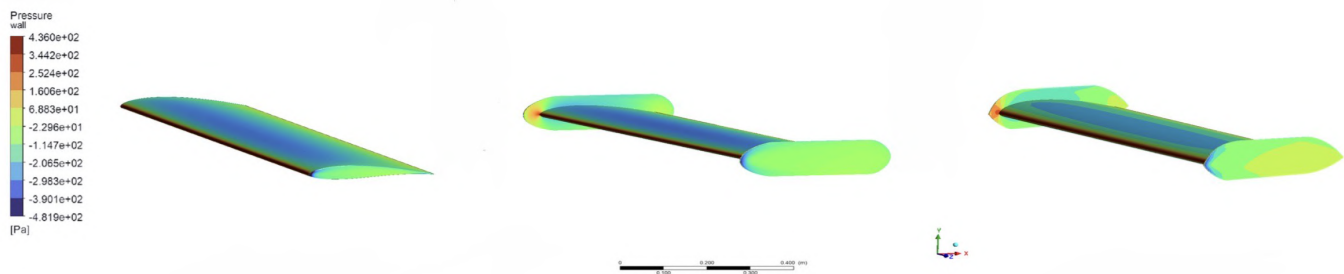


Figure 5.1.8: CFD pressure contours for endplate designs. Left to right: none, stadium endplate, airfoil endplate

5.1.6 Wing Integration Detailed Design

The wing integration piece is a 3D printed PLA structure used to securely attach the wing to the fuselage using two 6-32 screws through the spar, and two more in front of the spar. The top of the integration piece follows the shape

of the airfoil, while the bottom is flat to match the top of the fuselage. Three basswood half-ribs are used in order to minimize the weight contribution of the PLA. The CAD of the integration piece is shown in **Figure 5.1.9**.

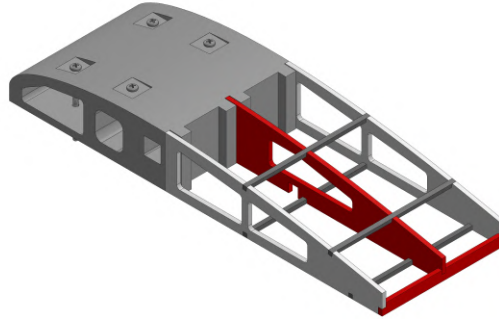


Figure 5.1.9: Integration piece CAD

5.1.7 Wing Assembly Analysis

To validate the structural integrity of the wing, FEA was conducted using ANSYS Static Structural on the full wing assembly. A force of **32 lbf**, representing a **3.5** load factor, was applied to the underside of the wing ribs, while a fixed displacement support was applied to the bottom of the integration piece. The results are shown in **Figure 5.1.10** and **Table 5.1.3**. The calculated FoS shows that the design is capable of withstanding the expected loads.

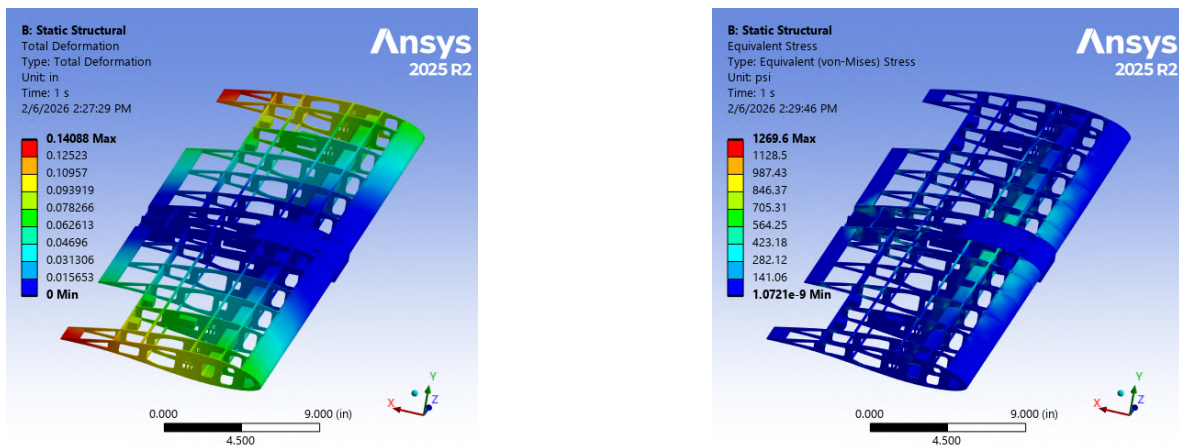


Figure 5.1.10: Wing Assembly deformation (left) stress distributions (right)

FoM	Basswood	CFRP
Max deflection (in)	0.14	0.08
Max stress (psi)	426.78	1269.60
FoS	20	236

Table 5.1.3: Wing assembly FEA results

5.1.8 Empennage Detailed Design

Both the vertical and horizontal stabilizers are designed as rib and stringer structures. The NACA 0012 airfoil was selected for both stabilizers, as it provides a symmetric profile with sufficient thickness to accommodate internal servo mounting while maintaining structural integrity.

The horizontal stabilizer includes a wooden box in the center to facilitate integration with the vertical stabilizer and the tail spar. The elevator utilizes the same triangular balsa trailing edge as the aileron in order to maintain the **2 degree** downward incidence angle. The rudder, on the other hand, uses two **1/32 in** balsa sheets to create a continuous, symmetric trailing edge. Consistent with the wing design, CA hinges and balsa reinforcement blocks are implemented, with PLA 3D printed leading edges added to the rudder and elevator. The final CAD renderings of the empennage configuration are shown in **Figure 5.1.11**.

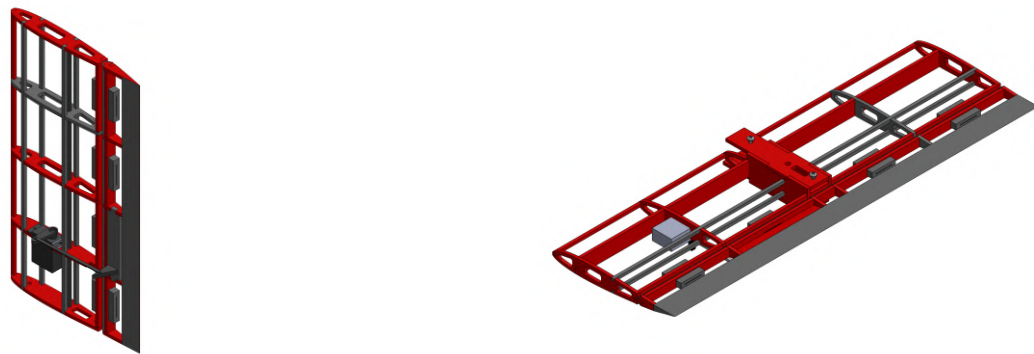


Figure 5.1.11: VT CAD (left) and HT CAD (right)

To validate empennage structural integrity, FEA was performed in ANSYS Static Structural using an **8 lbf** upward load applied to the underside of the horizontal stabilizer, representing a worst-case condition for aggressive maneuvering. Quantitative results are summarized in **Table 5.1.4**, with stress and deflection distributions shown in **Figure 5.1.12**.

FoM	Basswood	CFRP
Max deflection (in)	0.55	0.30
Max stress (psi)	2,800	13,510
FoS	3	22

Table 5.1.4: Empennage FEA results



Figure 5.1.12: Empennage deflection (left) and stress (right)

5.1.9 Propulsion and Avionics Detailed Design

A wiring schematic of the avionics layout is shown in **Figure 5.1.13**. The motor is mounted on the nosecone, with power leads routed through the forward bulkhead to the ESC, fuse, and battery in series. The fuse, located in the upper fuselage and discussed further in **Section 5.1.10**, remains accessible for inspection, while the battery is secured to the lower electronics bay using high-strength Velcro to ensure retention under flight loads.

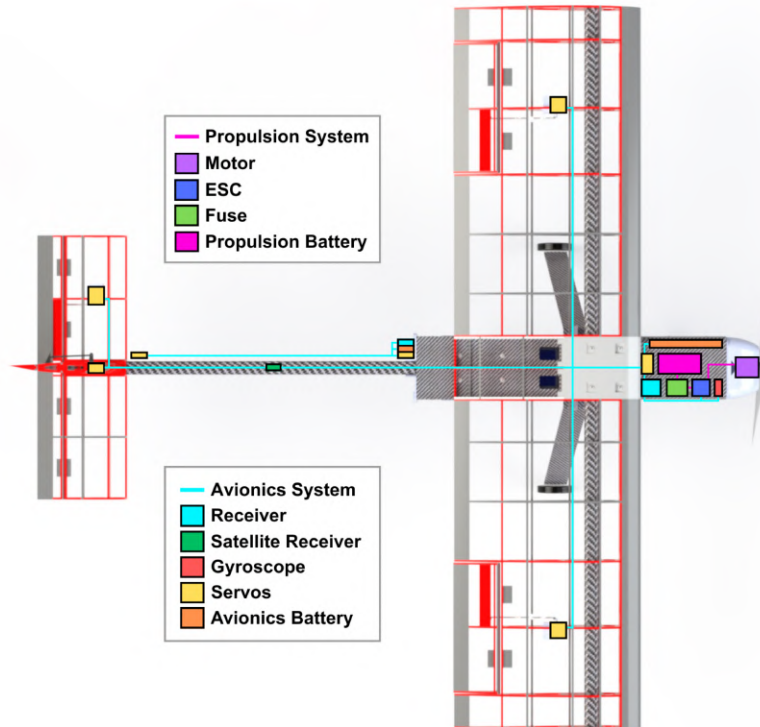


Figure 5.1.13: Wiring diagram of the propulsion system

As in **Table 5.1.5**, aircraft control for all missions is provided by a receiver–satellite pair, ensuring reliable signal reception and continuous pilot input throughout flight. Control authority is distributed across four flight control servos for aerodynamic surfaces and two dedicated release servos for mission mechanisms. An identical independent system is used for the banner release, connected to a separate transmitter.

Component	Product Specification
Receiver	Spektrum AR8020T DSMX 8
Satellite	Spektrum SRXL2 DSMX
Gyroscope	FMS Reflex V3
Control servo	Xpert CI-2401
Release servo	MKS-DS6100
Servo battery	URGENEX SCX24 900mAh 2S LiPo
Transmitter	Spektrum NX10+

Table 5.1.5: Avionic specifications

Control Surface	Required Torque (oz-in)
Aileron	25.01
Elevator	15.03
Rudder	8.96

Table 5.1.6: Servo requirements

The flight control servos were sized considering the available internal space and torque required to actuate the control surfaces at the estimated maximum airspeed of **140 ft/s**. **Equation 30** [14], was used to calculate the required

torque, which uses the reference dimensions of each control surface and a maximum deflection of **30 degree**. The results are summarized in **Table 5.1.6**, based on which the Xpert CI-2401 servo, with a rated torque of **83.56 oz-in** at **6V**, was selected for all control surfaces.

$$T_{\text{req}} = \frac{1}{2} \rho c^2 v_{\text{max}}^2 b \frac{\sin(\delta_{\text{max}}) \tan(\delta_{\text{max}})}{\tan(\delta_{\text{servo}})} \quad (30)$$

5.1.10 Avionics Integration Detailed Design

The avionics package is centered on a 3D printed PLA shelf mounted within the fuselage using pre-existing fuselage fastener locations. The fuse is retained in a 3D printed PLA holder epoxied to the shelf, constraining it in all directions. Placement directly beneath the fuselage access hatch ensures rapid access during testing and mission preparation. FEA was conducted in ANSYS Static Structural on the shelf and fuse holder assembly using a **20 lbf** insertion load determined using measurements. The final design was verified to operate with a FoS of **5.8**. The FEA analysis can be seen in **Figure 5.1.14**.

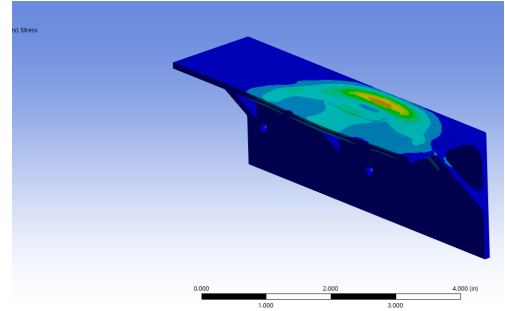


Figure 5.1.14: Avionics-shelf stress

5.1.11 Motor Integration Detailed Design

The motor is integrated into a 3D printed PLA nosecone mounted directly to the forward fuselage bulkhead. Relocating the cross-mount inside the fuselage allows the motor to clamp the bulkhead with four **M4** fasteners, eliminating a flat external interface and enabling the continuously curved forebody to experience less pressure drag.

Because the motor is enclosed within the nosecone, thermal performance was evaluated to ensure adequate cooling. A CFD analysis in ANSYS Fluent at M3 cruise conditions depicted in **Figure 5.1.15**, the lowest airflow case, predicted an average internal velocity of **26.25 ft/s** through the vented nosecone and motor housing. Using this velocity and the exposed motor area of **1.15 in²**, convective heat transfer calculations using **Equations 31** through **33** were performed assuming standard sea-level conditions and aluminum properties. The predicted steady-state motor temperature of **80°F** remained well below the PLA glass transition temperature of **140°F**, confirming thermal viability. Experimental validation of these results is presented in **Section 7.2.3**.

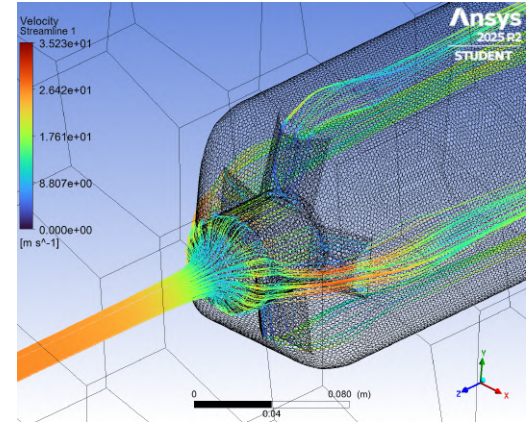


Figure 5.1.15: Nosecone airflow analysis

$$P_{\text{loss}} = I^2 R \quad (31)$$

$$h = \frac{k}{L_m} \left[0.3 + \frac{0.62 Re^{1/2} Pr^{1/3}}{\left(1 + \left(\frac{0.4}{Pr}\right)^{2/3}\right)^{1/4}} \left(1 + \left(\frac{Re}{2.86 \times 10^5}\right)^{5/8}\right)^{4/5} \right] \quad (32)$$

$$T_m = T_{\infty} + \frac{P_{\text{loss}}}{h A_m} \quad (33)$$

5.2 Mission Subsystem Design

5.2.1 Banner Detailed Design

In accordance with SB-R-01 and SB-R-02 requirements listed in **Table 3.1.1**, the banner design prioritized visibility and vertical orientation in flight. Three **0.03 lb** curtain weights were embedded along the lower edge of the banner, and an **1/8 in** CFRP stiffener was integrated into the leading edge. Fishing line was attached to the stiffener and positioned above the expected vertical CG to promote stable deployment. This configuration was later validated through testing described in **Section 7**.

To reduce aerodynamic drag, horizontal slits were incorporated along the banner length. Consistent with prior studies [7] and calculations presented in **Section 4.3.6**, the slits emulate the aerodynamic behavior of multiple high-aspect-ratio banners, resulting in a lower effective drag coefficient. The slits were loosely sewn to preserve overall banner geometry and ensure visibility of the Cornell University logo, which was applied using paint as shown in **Figure 5.2.1**.



Figure 5.2.1: Final banner design

5.2.2 Banner Integration Detailed Design

Stowed Banner Configuration The banner is stowed beneath the tail boom and is secured via two 3D printed PLA components: the deployment mechanism and the release mechanism. The weight of the banner is carried by an elastic band and a 3D printed PLA housing, one attached to each of the components. The housing and elastic band clamp the banner into position, preventing translation in all directions and ensuring the banner stays folded prior to deployment. **Figure 5.2.1** depicts the three phases of the banner flight.

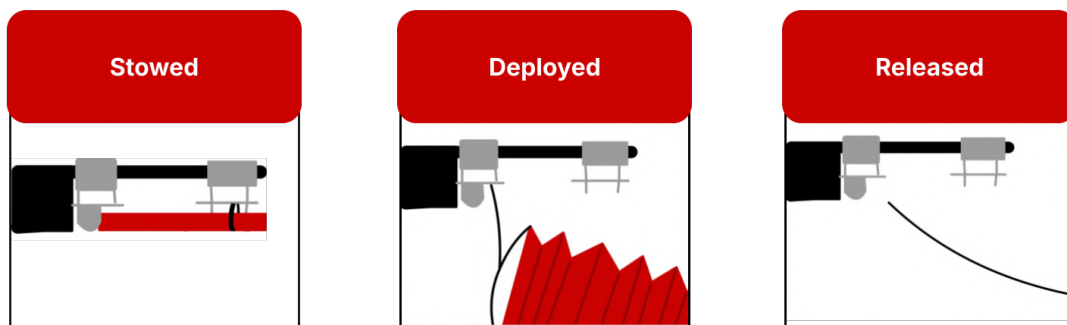


Figure 5.2.2: Banner stowing, deployment, and release

Deployment Mechanism The deployment mechanism consists of a 3D printed PLA assembly that attaches to the tail spar. The mechanism consists of three components: a spar-attachment clamp, a servo deployment module, and a spacer. The clamp is fabricated in two halves around the spar with each half incorporating heat set fabric snaps, allowing for secure and quick integration during GM. The servo deployment module and spacer serve to secure the banner while stowed and enable deployment in flight. The elastic band wraps around the spacer and banner. It is fixed to one end of the mechanism and loops around a pin attached to the servo, as shown in **Figure 5.2.3**. When signaled, the servo retracts the pin, and one end of the elastic band is released, allowing the banner to fall and deploy.

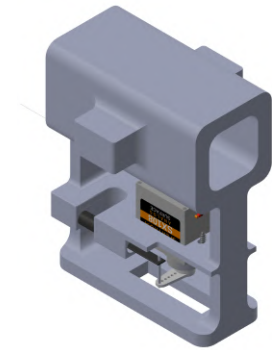


Figure 5.2.3: Banner deployment mechanism

Release Mechanism Similar to the deployment mechanism, the release mechanism consists of a 3D printed PLA assembly that attaches to the tail spar with fabric snaps. The release mechanism differs from the deployment mechanism as it also incorporates a housing for spooled towing rope and the stowed banner. Additionally, it houses the avionics system that controls and powers the two banner servos. Lastly, the servo pin restrains the towing rope instead of an elastic band, as shown in **Figure 5.2.4**. When the pin is extended, it provides a secure method for keeping the banner attached to the aircraft but also enabling it to rotate about the pin to remain streamlined. Upon the release command, the pin retracts, freeing the rope and allowing the banner to fall as prescribed.

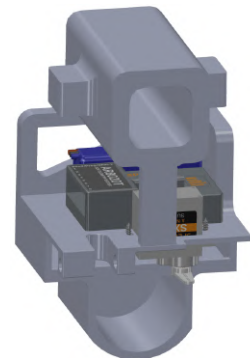


Figure 5.2.4: Banner release mechanism

Banner Placement Analysis To ensure the banner operates outside the aircraft wake, the downstream flow field of DF1 was analyzed using the steady $k - \omega$ SST model described in **Section 4.3.6**. Regions of elevated turbulence were identified using an isosurface of turbulence kinetic energy corresponding to 5% of the freestream mean kinetic energy [15], and velocity magnitude was plotted over this surface. Longitudinal midplane velocity contours were also extracted to assess wake structure. As shown in **Figure 5.2.5**, elevated turbulence dissipates shortly downstream of the empennage and does not extend significantly below the fuselage. Although a reduced-velocity wake persists aft of the aircraft, the absence of elevated turbulence indicates operation within comparatively uniform flow, supporting stable storage and release predominantly outside the wake.

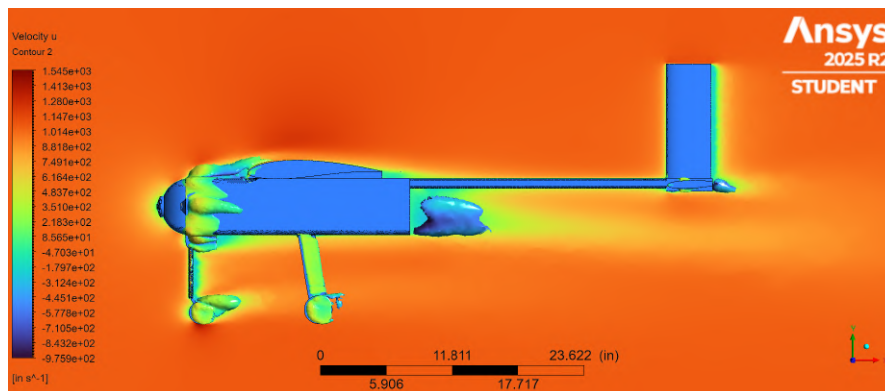


Figure 5.2.5: CFD velocity contours

5.2.3 Passenger Compartment Detailed Design

The passenger restraint system, shown in **Figure 5.2.6**, features a lightweight 3D printed mounting plate with guiding rails and an elastic latch assembly. The rails maintain passenger alignment, while a hinged upper panel locks into hook-style catches, enabling secure restraint and quick loading with minimal precision required. This design includes a PLA base and ABS topframe, both 3D printed. A latex exercise band is secured to the top frame using CA. The base and topframe are bound by two elastic connectors. When the topframe is latched, the elastic band conforms to the each passenger's unique shape to prevent translation in all directions.

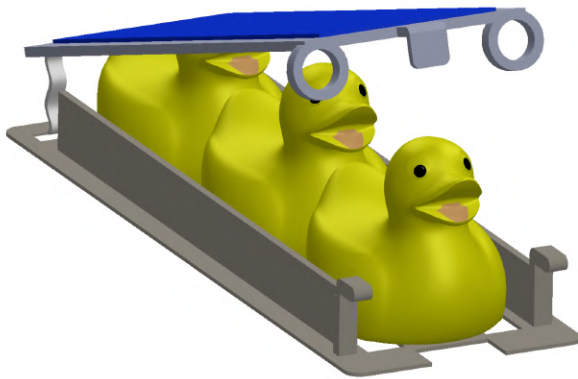


Figure 5.2.6: Passenger restraint CAD

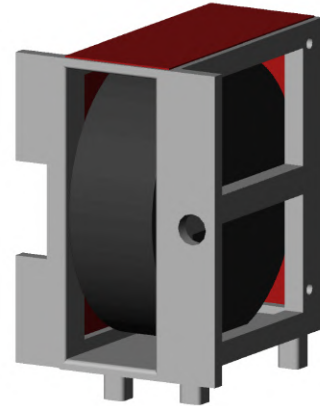


Figure 5.2.7: Cargo containment compartment CAD

5.2.4 Cargo Compartment Detailed Design

The cargo retention system, illustrated in **Figure 5.2.7**, is a 3D printed PLA five-sided housing designed to securely contain one vertically oriented cargo piece. A horizontally hinged hatch is incorporated into the side of the fuselage, which closes the compartment and prevents translation of the cargo, acting as a retention mechanism. To verify the cargo is secure, the hatch is fabricated from a transparent acrylic which allows visual confirmation. Without the cargo loaded, the system weighs **0.08 lbf**. To minimize weight, only the corners and edges of the box are fully formed, with MonoKote used to encapsulate the compartment completely. The box is secured vertically to the bulkhead inside of the fuselage by **2-56** nylon screws. This bulkhead also serves to encapsulate the final open side of the box.

5.3 Weight and Balance

The aircraft mass distribution varies between missions due to changes in payload configuration and external loads. To accommodate the resulting CG shift, the battery is repositioned within the electronics bay, where it serves as an effective ballast mass due to its significant contribution to total aircraft weight. For modeling purposes, the coordinate system origin is defined at the center of the wing leading edge, with the positive x-axis directed aft, the positive y-axis toward the right wing, and the positive z-axis upward.

Figure 5.3.1 shows the aircraft's mass distribution by functional subsystem for each mission profile. The mechanical subsystem refers to any supporting structural components that are not lifting surfaces and are not specific to missions. This includes the spars, the fuselage, the landing gear, and the nosecone.

Component	Weight (lbf)	x_{cg} (in)	y_{cg} (in)	z_{cg} (in)
Motor & Propeller	0.86	-8.74	0	-2.76
Nosecone	0.34	-6.34	0	-2.76
Front gear	0.23	-4.21	0	-8.28
Battery	1.52	-2.44	0	-2.76
Receiver & satellite receiver	0.05	-0.28	0	-2.76
Servo battery	0.10	-0.28	0	-2.76
Gyroscope	0.05	-0.28	0	-2.76
Wing integration & spar	0.74	3.50	0	0
Main wing	0.88	5.43	0	0
Main gear	0.28	4.72	0	-8.27
Fuselage	0.85	5.00	0	-2.77
Tail spar	0.36	16.34	0	0
Horizontal stabilizer	0.20	32.68	0	0
Vertical stabilizer	0.11	32.68	0	0
Mission 1				
Overall	6.55	2.32	0	-2.17
Mission 2				
Passenger	0.26	7.52	0	-2.76
Cargo	0.42	3.31	0	-2.76
Battery	1.52	-3.58	0	-2.76
Overall	7.23	2.43	0	-2.24
Mission 3				
Banner	0.51	20.39	0	-2.76
Banner Drop	0.30	20.39	0	0
Battery	1.52	-4.28	0	-2.76
Overall	7.37	3.33	0	-2.13

Table 5.3.1: Mission profiles total weight and distribution

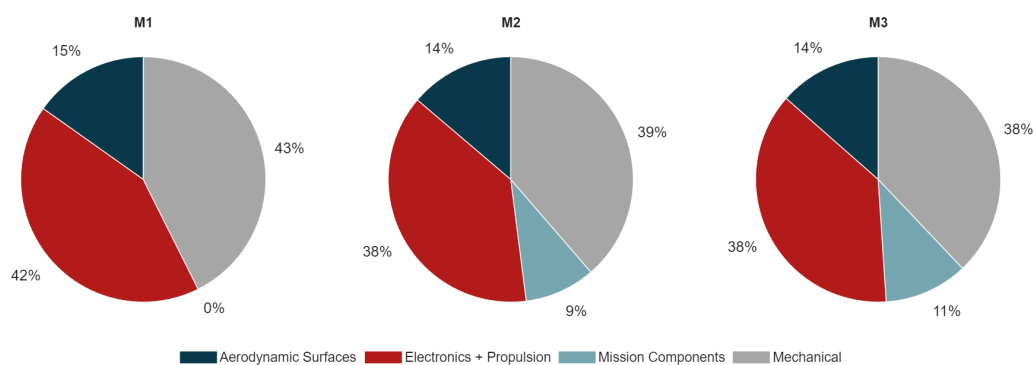


Figure 5.3.1: Mass breakdown for each mission

5.4 Aircraft Performance

Using the ANSYS Fluent polar curves generated in Section 4.4.7, key cruise velocities were identified. Maximum range and endurance conditions were determined by locating the peak C_L/C_D and $C_L^{3/2}/C_D$ ratios, yielding optimal lift coefficients of **0.354** and **0.601** for M1, respectively. Applying **Equation 26**, stall speed, maximum range speed, and maximum endurance speed were computed for each mission configuration. Additionally, the range and endurance was calculated for each mission at the cruise velocity as shown in **Figure 5.4.1**. The similarities between the cruise conditions of M1 and M2 allowed for a combined analysis of those two missions. M1 and M2 have a range of **6.20 mi** and an endurance of **6.00 min**. M3 has a range of **3.10 mi** and a range of **5.20 min**. These values are acceptable given the competition flight requirements.

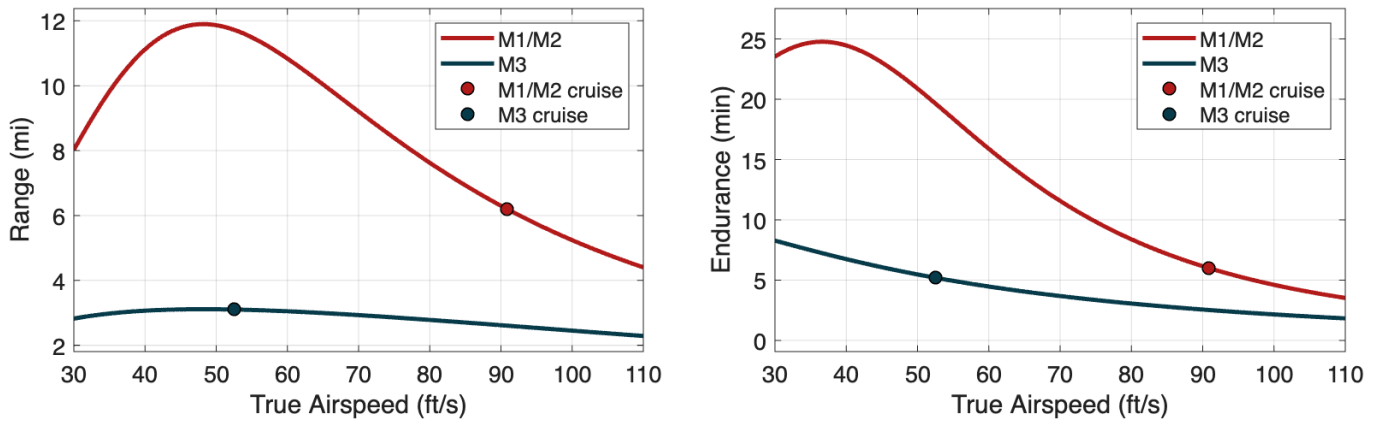


Figure 5.4.1: Range vs. airspeed (left) and endurance vs. airspeed (right) for M1

5.5 Mission Specific Performance

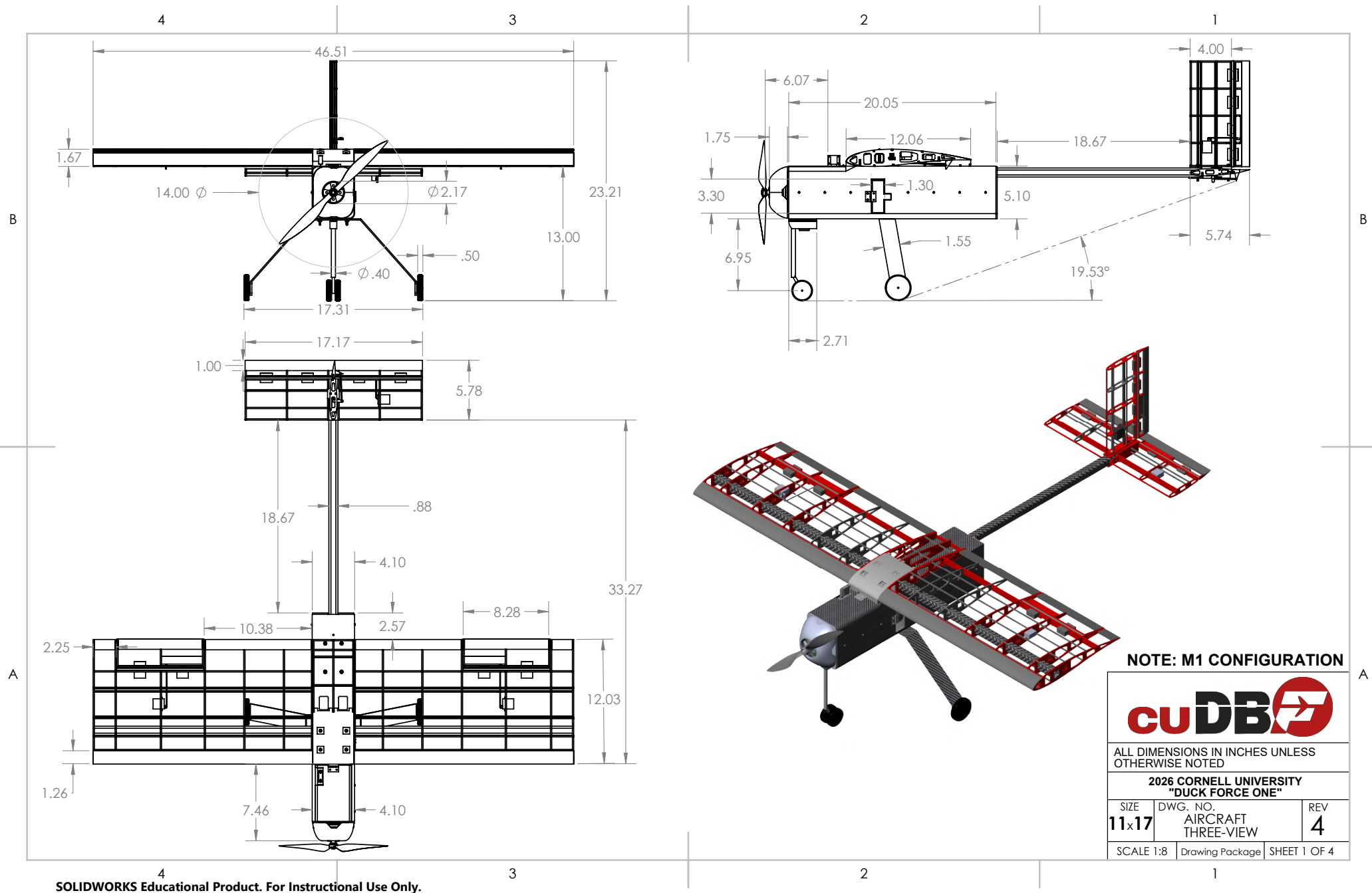
Using initial performance metrics estimated by **Section 4.3** genetic algorithm in conjunction with eCalc [4], XFLR5 [3], and OpenVSP [13], a final prediction of mission flight speeds, times, and scoring performance was constructed and compiled in **Table 5.5.1**.

Parameter	M1	M2	M3	GM
MTOW (lbs)	6.554	7.228	7.366	—
Cruise Velocity (ft/s)	90.88	90.88	52.48	—
Lap Time (s)	35.00	37.10	59.90	—
Number of Laps	3	8	5	—
Predicted Performance	Complete	140 profit	5 laps × 150 in	15 sec
Estimated Max Performance	Complete	2280	8 laps × 300 in	12
Mission Score	1.00	1.02	2.30	0.80
Total Competition Score	5.12			

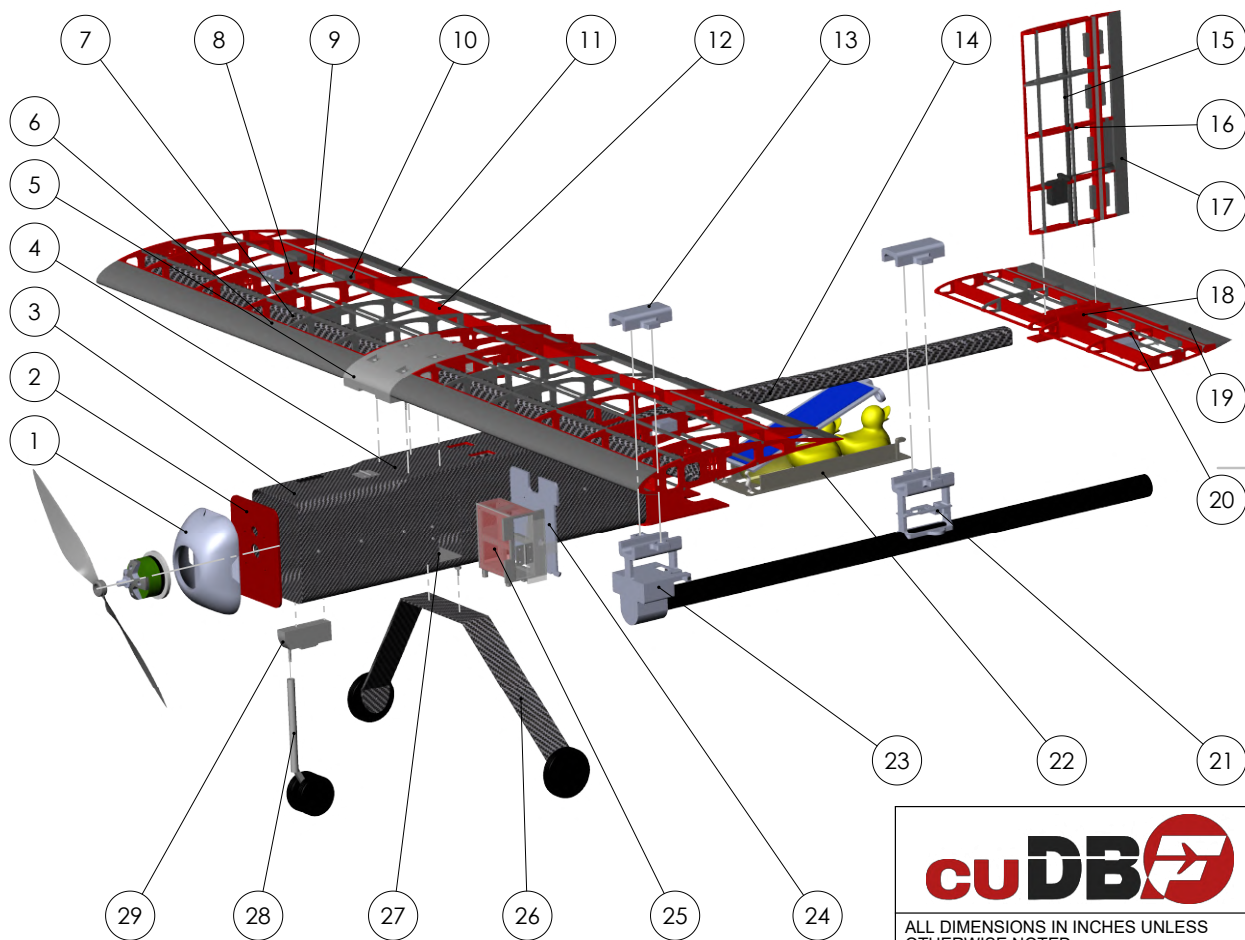
Table 5.5.1: Predicted mission-specific performances for final design

5.6 Drawing Package

The drawing package contains engineering drawings of the aircraft and its major subsystems. Included are 3-View, exploded view, and important individual components. Drawings have parts and major dimensions labeled and materials are assigned. All CAD models are created in SOLIDWORKS.



ITEM NO.	PART	MATERIAL	QTY
1	Nose Cone	PLA	1
2	Nose Mount Bulkhead	Basswood	1
3	Fuselage Monocoque	Carbon Fiber	1
4	Wing Attachment Plate	Basswood	1
5	Wing Integration Box	PLA	1
6	Wing Leading Edge Cover	Balsa	2
7	Wing Spar	Carbon Fiber	1
8	Wing Rib	Basswood/Balsa	12
9	Stringer	Balsa	11
10	CA Hinge Attachment Block	Balsa	12
11	Aileron	Basswood/Balsa	2
12	Wing Structural Comb	Basswood	2
13	Banner Mechanism Top	PLA	2
14	Tail Boom	Carbon Fiber	1
15	Vertical Stabilizer Stiffener	Carbon Fiber	1
16	Vertical Stabilizer	Basswood/Balsa	1
17	Rudder	Basswood/Balsa	1
18	Tail Integration Box	Basswood	1
19	Elevator	Basswood/Balsa	1
20	Horizontal Stabilizer	Basswood/Balsa	1
21	Banner Deployment Mechanism	PLA	1
22	Passenger Restraining Mechanism	PLA	1
23	Banner Release Mechanism	PLA	1
24	Passenger Compartment Bulkhead	Basswood	1
25	Cargo Restraining Mechanism	PLA	1
26	Main Landing Gear	Carbon Fiber	1
27	Main Landing Gear Attachment Plate	Aluminum	1
28	Front Landing Gear	Aluminum	1
29	Front Landing Gear Servo Housing	Aluminum	1





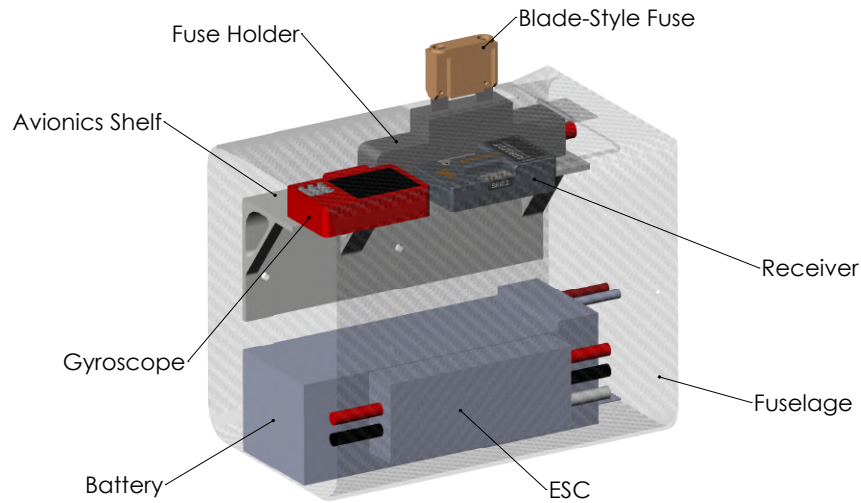
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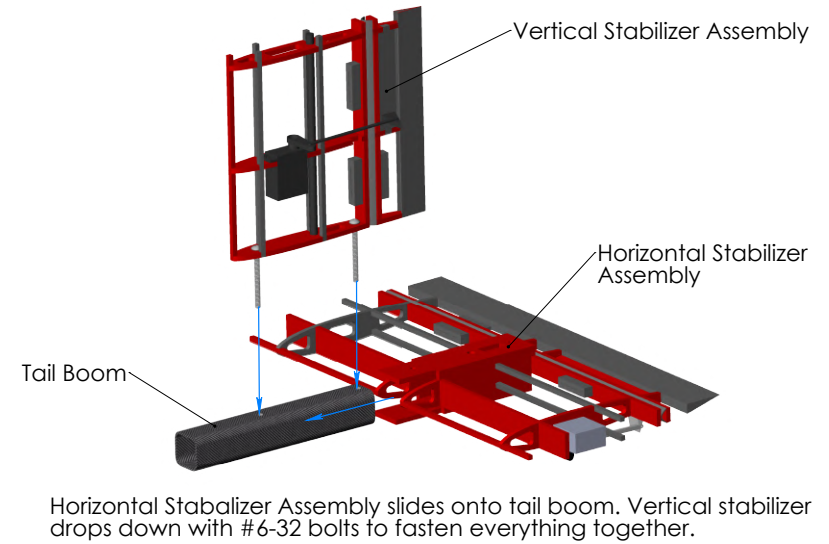
SIZE 11x17	DWG. NO. STRUCTURAL ARRANGEMENT	REV 4
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SCALE: 1:6 Drawing Package SHEET 2 OF 4

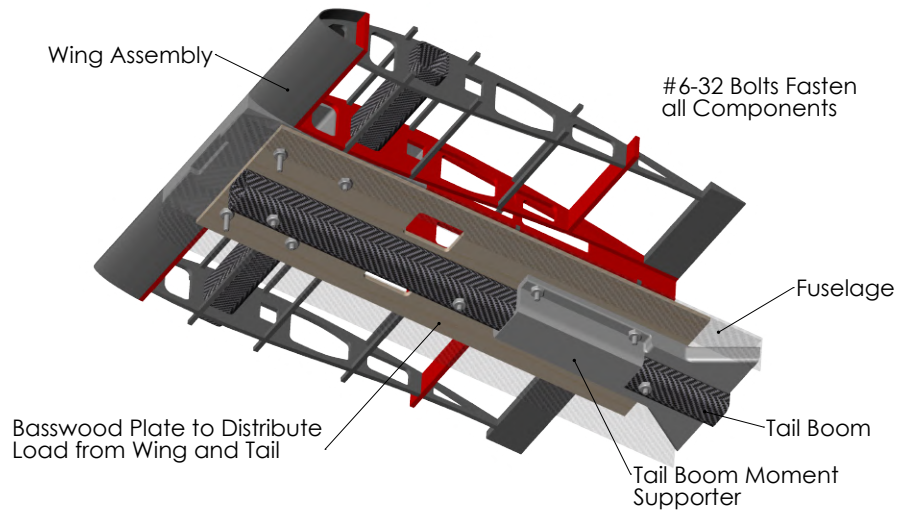
Avionics Bay Layout



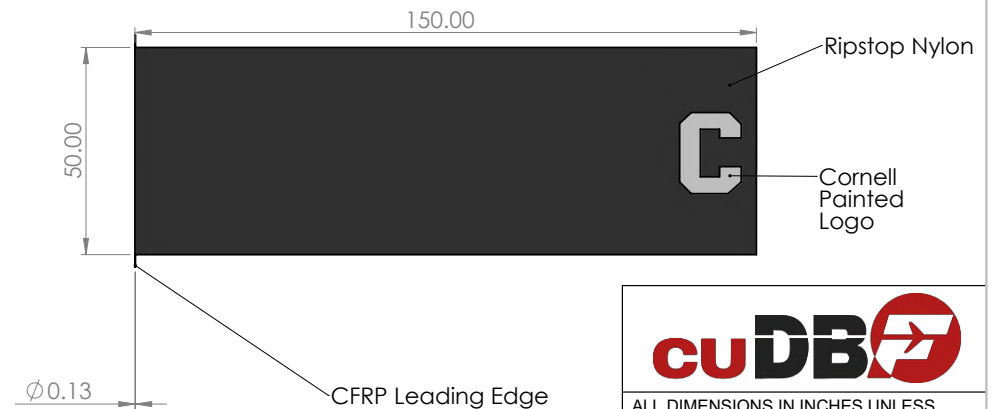
Empennage Integration (1:3)



Wing and Tail Boom Integration (1:3)

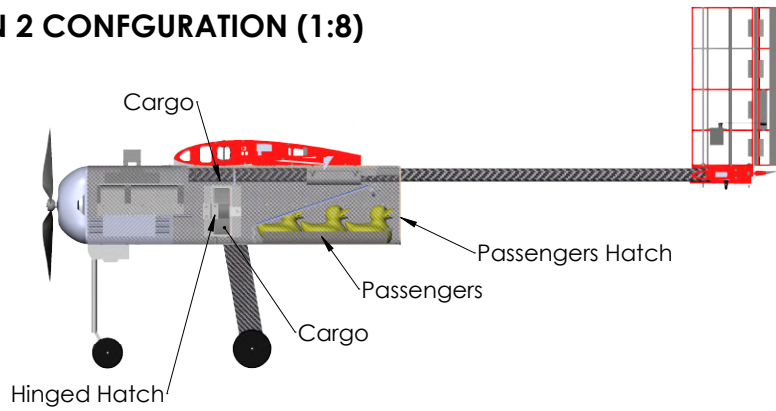


Banner Layout (1:30)

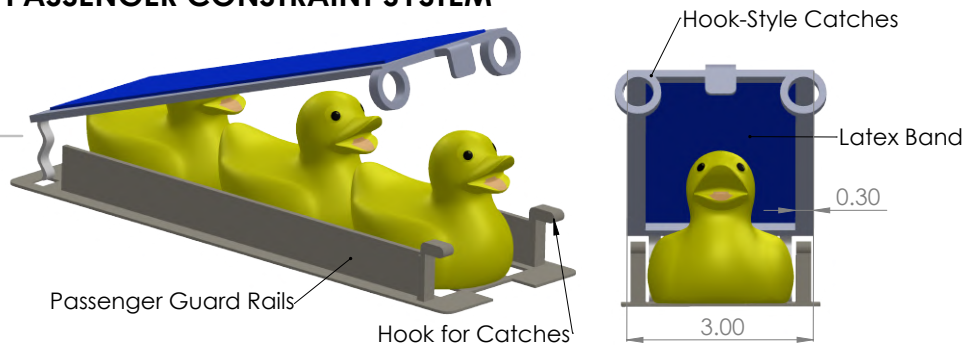


		
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SIZE 11x17	DWG. NO. SYSTEMS LAYOUT	REV 4
SCALE: 1:2	Drawing Package	SHEET 3 OF 4

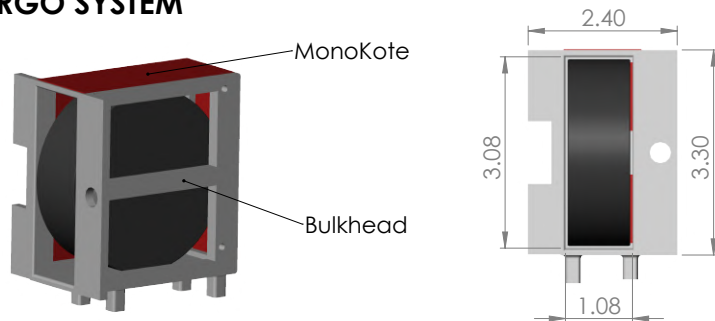
MISSION 2 CONFIGURATION (1:8)



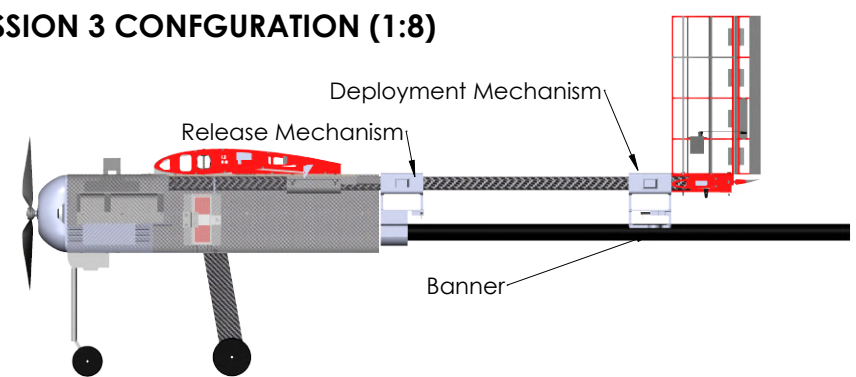
PASSENGER CONSTRAINT SYSTEM



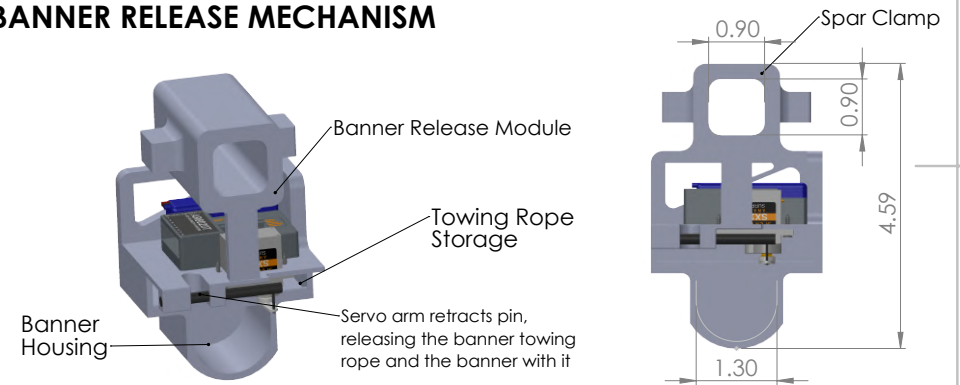
CARGO SYSTEM



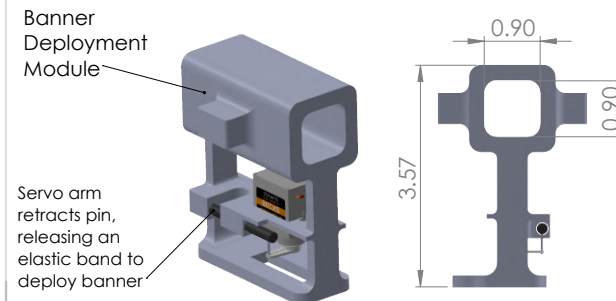
MISSION 3 CONFIGURATION (1:8)



BANNER RELEASE MECHANISM



BANNER DEPLOYMENT MECHANISM



		
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2026 CORNELL UNIVERSITY "DUCK FORCE ONE"		
SIZE 11x17	DWG. NO. MISSION CONFIGURATIONS	REV 4
SCALE: 1:2	Drawing Package	SHEET 4 OF 4

6 Manufacturing Plan

6.1 Manufacturing Process

The aircraft manufacturing process follows an iterative cycle with four phases: modeling and analysis, fabrication, system and subsystem assembly, and testing. Upon completion of each testing and analysis phase, the cycle was repeated for subsequent iterations. After sizing and scoring, the resulting configurations were developed in SOLIDWORKS [2] to assess component function and integration prior to fabrication. Once finalized, designs progressed to manufacturing, where fabricated and COTS components were assembled. Identified integration issues prompted revisions and additional fabrication cycles as needed. All systems and subsystems were then subjected to the testing outlined in **Section 7**, with results used to verify performance and determine whether further redesign were required.

6.1.1 Additive Manufacturing

Additive manufacturing is inexpensive, fast, and enables the fabrication of components with complex or precise geometry. CUDBF has access to the Cornell University RPL, which houses several Bambu Lab X-1 Carbon 3D printers that can be loaded with materials such as PLA and ABS. The printers were utilized in producing various components such as the banner release system and CFRP molds.

6.1.2 Laser Cutting

A laser cutter was used to precisely fabricate balsa and basswood designs, especially prevalent in the wing and empennage. The Epilog Laser Fusion Edge, housed in the RPL, was used to produce these components such as airfoils with accuracy, enabling the team to efficiently fabricate multiple identical parts.

6.1.3 Composite Layups

Wet carbon fiber composite layups are used to manufacture structural components with complex custom components such as the fuselage. Impregnated CFRP sheets are layered with a peel ply release film, a perforated release film, and fabric breathers in 3D-printed molds. The filled molds are left to cure in a vacuum bag for 18 hours. Following release from the molds, the components are sanded and cut using a Dremel rotary tool to smooth out any imperfections. The use of pre-impregnated carbon fiber (prepreg) was investigated by CUDBF for its superior ability to create complex shapes with smooth edges versus standard wet layups. However, prepreg presents several complications including requiring baking at high temperatures, making 3D printed molds impossible, and storage at freezing temperatures. The team decided that more off season research and development is necessary before committing to the use of prepreg on the competition aircraft.

6.1.4 Adhesives

The manufacturing of DF1 employs four main adhesives: CA, wood glue, epoxy resin, and Velcro. CA is a fast-bonding adhesive used for permanent balsa-to-balsa bonds, such as between ribs and stringers. Wood glue provides stronger adhesion for tightly bonded wood interfaces. Epoxy is used to attach plastic and composite components to each other. Velcro serves as a temporary adhesive for repositionable components, such as the battery and receiver.

6.1.5 Monokote Covering

MonoKote is a lightweight, heat-activated plastic film used to cover aircraft surfaces. When applied, heat causes the film to shrink and conform to the shape of the underlying structure, creating a smooth aerodynamic surface. This process provides a strong, lightweight finish that improves airflow over wing, tail, and control surfaces, helping to reduce drag and flow separation.

6.2 Subsystem Manufacturing

6.2.1 Fuselage Manufacturing

DF1's fuselage was manufactured via a wet layup, shown in **Figure 6.2.1**, as two CFRP clamshell halves shaped by 3D printed molds. After curing, halves were removed from the molds, trimmed, and machined to create hatches and fasteners, with 3D printed alignment fixtures used to match hole locations between halves. Three laser-cut **1/8 in** basswood bulkheads and an internal wing mounting plate were bonded in place with epoxy to provide compartmentalization and structural support, separating avionics, passenger, and cargo spaces per competition requirements.



Figure 6.2.1: Fuselage fabrication (left and middle) and post processing (right)

6.2.2 Wing Manufacturing

The wing and ailerons are wooden rib and stringer structures built using wood glue and CA, and mounted on a square carbon fiber spar. The ribs and combs were laser cut from **1/8 in** balsa or basswood, depending on whether the component served a load-bearing or primarily geometric function. The wood structure before applying MonoKote is shown in **Figure 6.2.2**. The leading edge of the wing was formed from **1/32 in** balsa sheet, which was soaked in water and molded to the airfoil contour. The trailing edge **1/8 in** balsa stringers were bonded into rib notches using CA to provide a supporting skeleton for the MonoKote.

The servos are mounted to a wing rib using **2-56** nylon fasteners. The wing halves and ailerons were covered with MonoKote and are attached together using CA hinges.

6.2.3 Empennage Manufacturing

Consistent with wing manufacturing, the stabilizers and their respective control surfaces are rib and stringer structures built with laser cut **1/8 in** basswood and balsa wood, and bonded using CA and wood glue. **Figures 6.2.3** displays the wooden structures of the empennage. The stabilizers and their respective control surfaces were covered with MonoKote separately and then attached to each other via the CA hinges. To ensure accurate rib alignment and

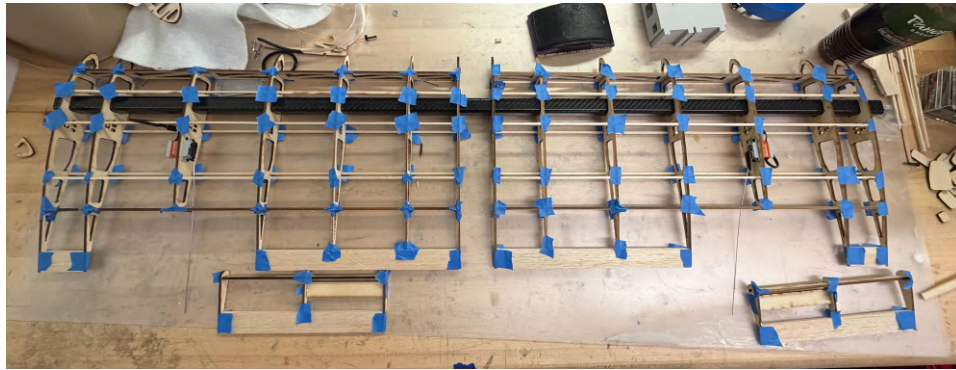


Figure 6.2.2: Wing and aileron wooden structures

placement, alignment rigs matching the stabilizers' dimensions were 3D printed to aid the construction process, as illustrated in **Figure 6.2.4**. To integrate the empennage, the tail spar is first inserted into the integration box at the center of the horizontal stabilizer. The vertical stabilizer is then positioned on top of the horizontal stabilizer and secured with steel **6-32** screws all the way down through the spar. Empennage servo wires are routed into the spar during assembly.

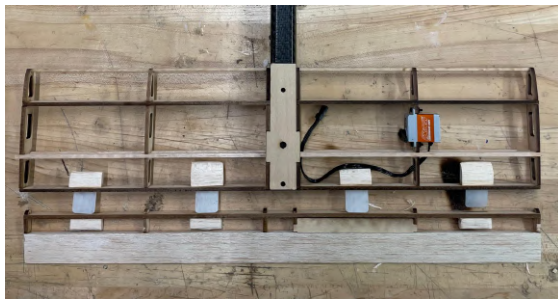


Figure 6.2.3: HT (left) and VT (right) wooden structures

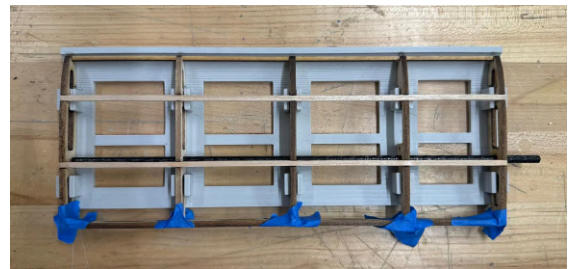
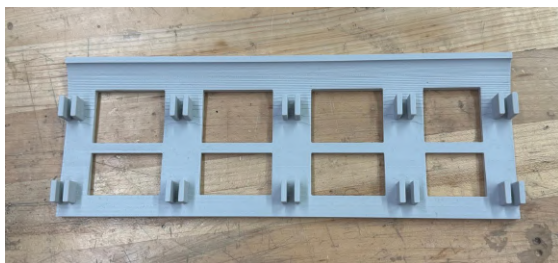


Figure 6.2.4: Tail alignment rig used for the VT

6.2.4 Mission Packages Manufacturing

Passengers The passenger restraint system was designed in SOLIDWORKS and fabricated from PLA via 3D printing. An elastic exercise band, cut to length, was bonded across the upper frame using CA to provide compliant

restraint. The upper and lower portions of the assembly are retained using elastic bands, and the completed passenger module was bonded directly to the fuselage using epoxy.

Cargo The cargo retention structure was also designed in SOLIDWORKS and 3D printed in PLA. Open faces of the structure were covered with MonoKote to retain the cargo while minimizing added mass. The completed cargo module was installed in the fuselage using epoxy to ensure security under flight and landing loads.

Banner The banner was fabricated from ripstop nylon cut to the required dimensions. A carbon fiber tube serves as the banner leading edge and was bonded to the fabric using CA. Sewn pockets along the trailing edge accommodate removable ballast weights. Two custom PLA attachment components were designed to interface the banner with the tail spar. One attachment secures the banner forward of the tail boom, while the second attaches the aft end at the tail spar tip.

6.3 Manufacturing Milestones

The team uses a manufacturing Gantt chart to keep track of important milestones during the manufacturing phase. A chart is made for each iteration, with each subteam keeping track of the anticipated and actual manufacturing process of different components. **Figure 6.3.1** shows the chart used for the 2025-2026 DBF cycle. The team plans to complete three manufacturing iterations prior to finalizing the competition aircraft, providing multiple opportunities to identify design or integration issues and implement solutions. Manufacturing activities are structured to occur in parallel where possible in order to accelerate the build process and maximize productivity. The final week of each build phase is dedicated to subsystem integration and assembly verification to confirm that all components function together as intended. An additional buffer period is incorporated at the conclusion of the competition aircraft build to accommodate unforeseen delays.

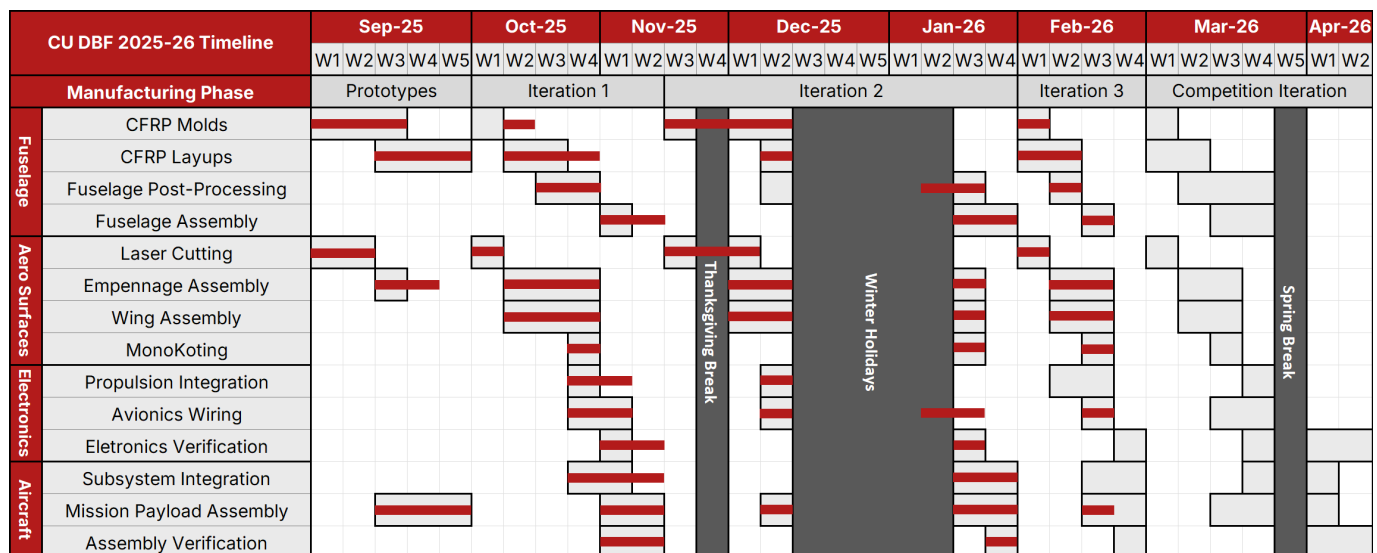


Figure 6.3.1: DF1 manufacturing schedule

7 Testing Plan

7.1 Testing Schedule and Objectives

To ensure that DF1 and its subsystems would perform under design constraints and fulfill competition mission objectives, testing was conducted to verify functionality, reliability, and performance. Testing objectives and schedule are found below in **Table 7.1.1** and **Figure 7.1.1** respectively.

Test	Objective	Method	Subteam
Ground Tests			
Wing Spar Loading Test	Validate that the wing spar can withstand the expected wing loading	Mount wing spar securely and apply predicted force at the tips	Mechanical & Aerodynamics
Landing Gear Load Test	Ensure the landing gear and fuselage can support the expected 3.5 load factor	Drop the aircraft loaded at MTOW from short heights and inspect for damage	Mechanical
Ground Mission Test	Predict and improve GM time	Perform and time the GM sequence; iterate and practice	Mechanical
Static Thrust Test	Verify and refine propeller choice; compare results with eCalc and MATLAB predictions	Use in-house static thrust test rig to measure generated static thrust	Propulsion
Endurance Test	Ensure electronics last the required mission lengths	Run motor at max throttle until battery is drained	Propulsion
Wind Tunnel Nosecone Test	Evaluate aerodynamic and thermal performance of the nosecone	Simulate cruise in wind tunnel to measure drag and motor temperature	Propulsion & Aerodynamics
Wind Tunnel Banner Test	Determine which banner material produces the least drag	Measure drag generated by banners of different materials	Mechanical & Aerodynamics
Banner Release Strength Test	Validate strength of fabric snap clamps under banner towing loads	Use force gauge to pull release mechanism until failure	Mechanical
Flight Tests			
Performance Test	Find maximum, cruise, and stall speeds	Fly aircraft and collect pitot-tube airspeed and flight controller data	Propulsion
Banner Release Test	Ensure banner releases properly and unfurls	Actuate banner release servos and unfurl banner mid-flight	Mechanical & Propulsion

Table 7.1.1: Testing Plan

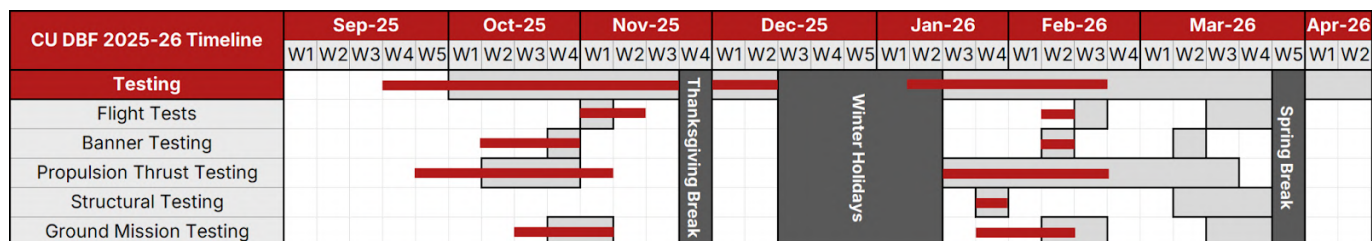


Figure 7.1.1: Testing Schedule

7.2 Ground Tests

7.2.1 Load and Drop Tests

Wing Spar Loading Test The CFRP wing spar was secured at its center to the edge of a testing platform using a clamp. The end of the spar was pulled upward using a force gauge until the target load was reached, demonstrated in **Figure 7.1.2**. During the test, the spar was monitored for visible failures, and the vertical deflection was from the unloaded configuration to the maximum applied load. These results were compared to deflections predicted by FEA.

Landing Gear Load Test The landing gear underwent a drop test to verify its structural integrity and attachment to the fuselage. DF1, in its MTOW configuration, was dropped from a height of **8.5 in** to replicate a **6.75 ft/s** descent rate. This descent rate was determined as a



Figure 7.1.2: Spar loading test

worst case estimate derived from historical team data and kitplane flight computer data. Following the drop, both the landing gear and fuselage were inspected for visible damage and structural failure.

7.2.2 Ground Mission Testing

GM Test GM testing was performed over the course of several iterations of the aircraft. This process ensured continued improvement of GM time with every iteration, since DF1 is optimized for a high-scoring fast GM time. A full-scale foam model of the DF1 fuselage was fabricated to function as the GMTV, as shown in **Figure 7.2.1**. The GMTV was used for testing when the aircraft was unavailable or incomplete. To ensure an accurate simulation of the mission, a **10 ft by 10 ft** floor section was marked off using tape and DF1 was placed inside the box; a tape mark was also made **1 ft** away from the box so the crew member could return to the same starting position for each test. GM testing, shown in **Figure 7.2.2**, was then executed as the **Section 3.1.3** rules specified. The test was conducted with multiple members to eliminate human error as a possible contributor to outlier data. After each test, a brief discussion of strategic improvements occurred and were implemented in the next test.



Figure 7.2.1: GMTV



Figure 7.2.2: DF1 GM testing

7.2.3 Live Propulsion Tests

Testing was conducted on a custom-built static thrust rig constructed of wood, shown in **Figure 7.2.3**. The motor, nosecone, and propeller were mounted to an upper beam connected to a lower beam via rigid supports, forming a pivoting lever arm. The lower beam applied force to the load cell as thrust generated a moment about the pivot. Thrust was logged via an Arduino, while current, voltage, throttle, and power were recorded using a power analyzer integrated into the propulsion power line.

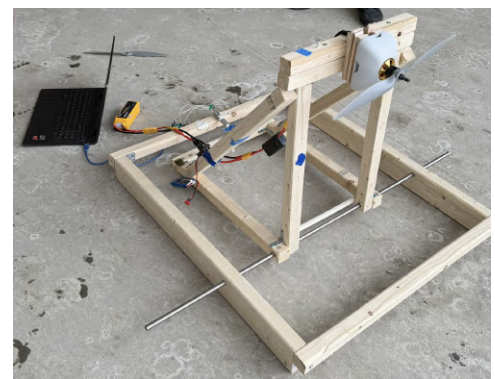


Figure 7.2.3: Static testing rig

Static Thrust Test Static thrust testing was conducted to validate propulsion performance predictions and compare candidate propulsion configurations. Testing focused on quantifying thrust output and electrical performance across multiple propeller, battery, and motor combinations using a controlled static test setup. Four propellers identified in **Section 4.3.4 (15 in × 6 in, 14 in × 8.5 in, 14 in × 7 in, and 13 in × 10 in)** were evaluated in an initial screening, from which the two highest-thrust configurations were selected for further testing. Subsequent tests compared battery and motor combinations using these propellers to assess tradeoffs between thrust, efficiency, and endurance, and to benchmark

measured performance against eCalc [4] and MATLAB predictions [11]. These comparisons were used to identify the propulsion configuration best suited to meeting mission requirements.

Endurance Test Endurance testing was conducted to verify that each propulsion configuration satisfied required mission flight durations. For each configuration, the motor was operated at full throttle, and the time a fully charged battery could sustain maximum load was recorded.

Wind Tunnel Nosecone Test Two tests were performed to evaluate the aerodynamic and thermal performance of the nosecone configuration. A static full-discharge test was conducted to assess thermal behavior, with motor temperature measured using an infrared thermometer at the conclusion of the run. Additionally, wind tunnel testing was performed with the motor operating at maximum RPM while freestream velocity was incrementally increased. As shown in **Figure 7.2.4**, the nosecone assembly was mounted to a load cell to quantify aerodynamic drag under cruise conditions.

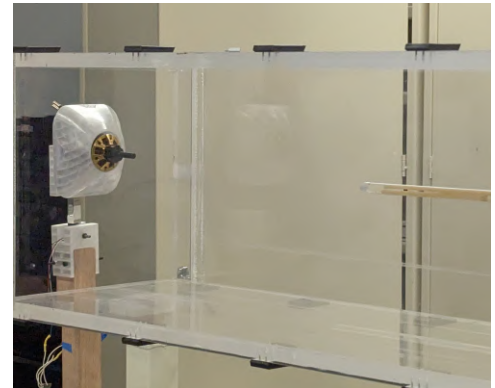


Figure 7.2.4: Wind tunnel setup

7.2.4 Banner Testing

Wind Tunnel Banner Test The aerodynamic performance of the banner was evaluated to quantify drag, assess deployment behavior, and establish structural and attachment requirements. Testing consisted of a combination of small-scale wind tunnel experiments and scale aircraft flights.

Wind tunnel testing was conducted in the Cornell University Bellis Fluid Dynamics Lab to inform material selection for the banner. Ripstop nylon and LDPE were evaluated. Miniature banners measuring **5 in x 1 in** were mounted to a rigid rod and placed in the wind tunnel as seen in **Figure 7.2.5**, with the banner assembly attached to a load cell to measure aerodynamic forces directly. Force data recorded from the load cell were used to estimate skin-friction drag to compare relative aerodynamic performance between materials. Following material selection, a scale banner was affixed to a COTS RC aircraft to test it under realistic flight conditions. The banner was scaled by the relationship in propulsion systems between the test aircraft and DF1.

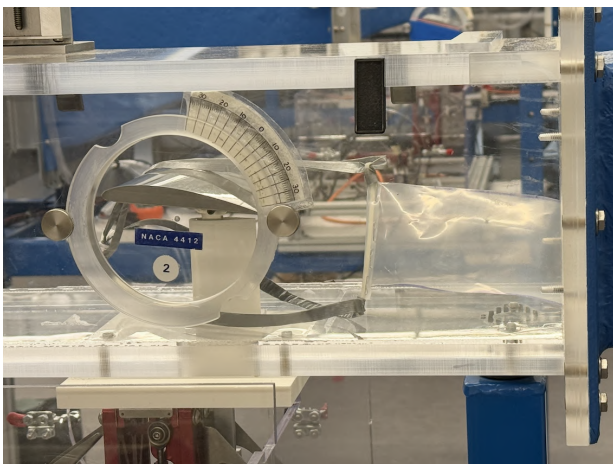


Figure 7.2.5: Wind tunnel banner testing



Figure 7.2.6: Banner pull test

Banner Release Strength Test To ensure that the banner deployment and release mechanism is capable of withstanding the expected loading from the banner pulling on it, a strength test was conducted on the banner release part of the mechanism. As shown in **Figure 7.2.6**, the release mechanism was attached to a spar and hooked to a force gauge. The gauge was pulled at a **45 degree** angle, the expected worst-case load path, until the snaps detached the piece from the spar.

7.3 Flight Testing Plan

A flight test program was conducted with support from experienced RC pilots at the Ithaca Radio Control Society to evaluate aircraft performance and identify areas for refinement. Each test sequence began with a standardized preflight inspection detailed in **Table 7.4.1**. Several flights were conducted each testing day, and the preflight checks were repeated before each one. Aircraft performance was evaluated through pilot feedback and data analysis, with emphasis on handling through a simulated mission course. Onboard instrumentation was used to collect cruise power and speed data for comparison against predicted values. The test results, described in **Section 8**, guided subsequent design refinement in each future iteration.

7.4 Flight Test Checklist

Ground Inspection Checklist		Preflight Checklist	
Category	Item	Category	Item
Assembly	Aircraft skin tear-free	Power	Propulsion battery full charge
	Wing & tail connection		Receiver battery full charge
	Control surfaces linkages		Transmitter battery full charge
	Landing gear		Transmitter Range test
Propulsion	Motor secure	Tests	Landing Gear / Roll test
	Propeller securely fastened		Propulsion Mount Structural Test
	Propeller direction		Wingtip test
Avionics	Ground clearance	Final checks	50% Throttle
	Propulsion battery secured		Throttle Closed
	Servo battery secured		Full Up Elevator
	Receiver & satellite receiver		Full Right Rudder
	Zero servos		Full Right Aileron
	Servo connections to receiver, battery		Radio Failsafe set
Actuation	Controls have enough deflection	Post-Flight Checklist	
	Control surfaces resist load	Fuse removed	
	Banner Mechanisms function	Visual Inspection	

Table 7.4.1: Flight test checklist

8 Performance Results

8.1 Subsystem Test Results

8.1.1 Load and Drop Test Results

Wing Spar Loading Test During testing, the wing spar exhibited no visible damage and returned to its original configuration once the load was removed. The wing spar was tested to a load of **26.70 lb** and a maximum deflection of **0.94 in** was measured. This was significantly lower than the **10 in** deflection predicted by FEA and was therefore considered negligible.

Landing Gear Load Test After being dropped from a height of **8.50 ft**, the landing gear and fuselage were inspected for structural damage. No failures were observed in either component as pictured in **Figure 8.1.1**. The aircraft remained upright after impact, and the landing gear attachment points showed no signs of shear or loosening, confirming the stability of the system under simulated landing loads.

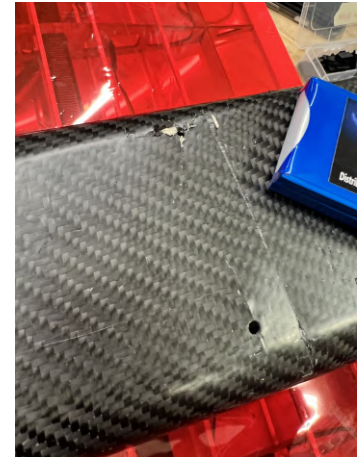


Figure 8.1.1: Landing gear test impact

8.1.2 Ground Mission Testing Results

GM Test The times collected from the GM simulation tests are collected in **Table 8.1.1**. Testing saw the average GM time decrease from **35.21 sec** to **15.37 sec** between the original mission prototypes and the second iteration of DF1. The results of this testing confirmed that successive iterations of the mission components improved GM times and thus GM score. Overall, the testing demonstrated a convergence to the GM time found by the MDO model described in **Section 4.2**.

Iteration of DF1	Prototypes	Iteration 1	Iteration 1.5	Iteration 2
Average GM Time (s)	35.21	22.36	20.65	15.27

Table 8.1.1: Simulated GM results over time

8.1.3 Propulsion Test Results

Static thrust testing conducted with the propellers outlined in **Section 4.2** is illustrated in **Figure 8.1.2**, which presents thrust as a function of throttle setting for each propeller evaluated. Based on these results, the **15 in x 6 in** and **14 in x 7 in** propellers were selected for further testing. As shown in **Figure 8.1.2**, the measured thrust for the **15 in x 6 in**, was plotted for both the **4200 mAh 35C** and **4000 mAh 30C** batteries. The corresponding eCalc [4] and MATLAB predictions [11] were consistently underperformed.

Comparison of the two final propeller configurations shows a clear tradeoff between thrust and endurance. Although the **14 in x 7 in** propeller produced lower maximum thrust, it drew less current and therefore offered a longer estimated flight time. In contrast, the **15 in x 6 in** propeller produced higher thrust but at the cost of increased current draw, reducing flight endurance. Further, the propeller-specific studies revealed that across all test configurations, the **4200 mAh** battery consistently produced higher maximum thrust than the **4000 mAh** battery. **Table 8.1.2** summarizes the measured maximum thrust, voltage, current, and power values under different combinations.

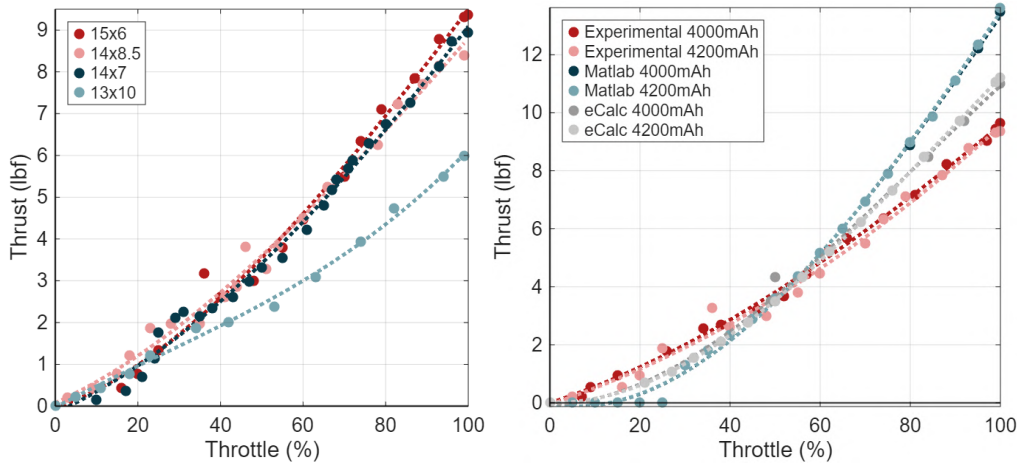


Figure 8.1.2: Initial propeller thrust test data (left) and 15 in x 6 in thrust data (right)

Propeller	Battery	Current (A)	Voltage (V)	Power (W)	Max Thrust (lbf)
14 in x 7 in	4200 mAh – 35C	52.71	22.04	1162	8.969
	4000 mAh – 30C	44.74	20.53	918.5	8.426
15 in x 6 in	4200 mAh – 35C	60.39	22.48	1358	10.34
	4000 mAh – 30C	52.14	20.63	1075	9.800

Table 8.1.2: Performance data from propeller-battery configurations

8.1.4 Endurance Test Results

Endurance testing confirmed that all four propulsion configurations met the required maximum-throttle run time. Results, summarized in **Figure 8.1.3**, revealed two primary trends. First, the **4000 mAh** battery, with its lower C-rating, exhibited a slower transient response to maximum thrust for both propellers; however, after approximately **90 sec**, thrust output converged for both battery options. Second, thrust decreased steadily over the duration of each test as available battery energy was depleted. The **15 in** propeller with the **4200 mAh** battery provided an increase of approximately **20 sec** of runtime. This trend was not observed with the **14 in** propeller, where the lower-capacity **4000 mAh** battery achieved approximately **30 sec** longer endurance. This behavior is attributed to the slower spool-up associated with the lower C-rating, reducing initial power draw and extending overall runtime.

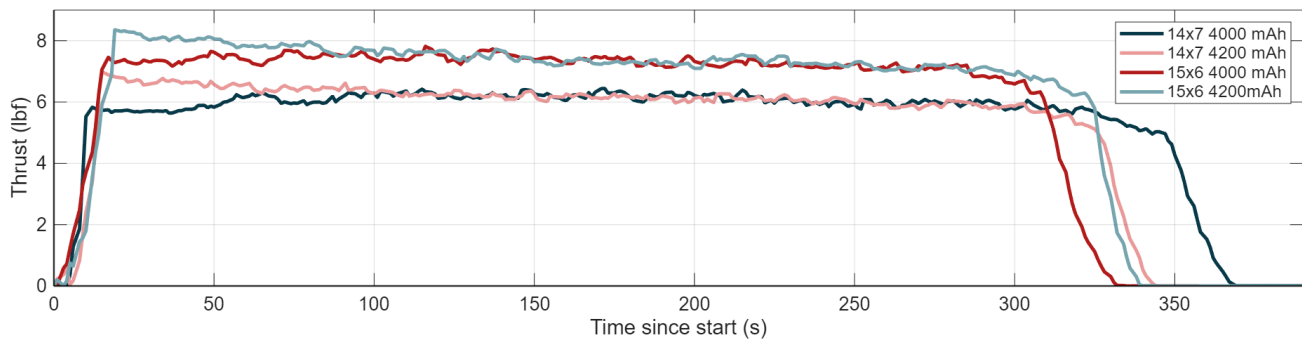


Figure 8.1.3: Max thrust over time until battery depletion

8.1.5 Nosecone Test Results

The thermal nosecone testing revealed a steady-state value of **90** at the end of the discharge, which lasted for approximately **300 sec**. This value reached **10** higher than the value predicted in **Section 8.1.5**, but still remained significantly below the glass transition temperature of PLA with a FoS of **1.15**.

The drag testing, illustrated in **Figure 8.1.4**, captured a nosecone drag coefficient **0.7**. Using **Equation 34** to scale the measured nosecone drag coefficient by the wing reference area, the resulting aircraft drag contribution was computed as **0.251**.

$$C_{D_{DF1}} = C_{D_{Nosecone}} \times \frac{A_{Nosecone}}{S_w} \quad (34)$$

8.1.6 Banner Test Results

Aerodynamic Testing Results Individual trial measurements from wind tunnel testing are summarized in **Table 8.1.3**. Ripstop nylon exhibited significantly lower drag and was therefore selected for the final banner design.

Scale RC aircraft testing showed that overall material performance trends were consistent with wind tunnel testing. Initial tests showed that the banner failed to remain upright so a leading-edge stiffener in the form of a carbon fiber rod was implemented to suppress excessive flutter and maintain banner shape. Testing with the rod proved it to be a viable technique for keeping the banner vertical in flight.

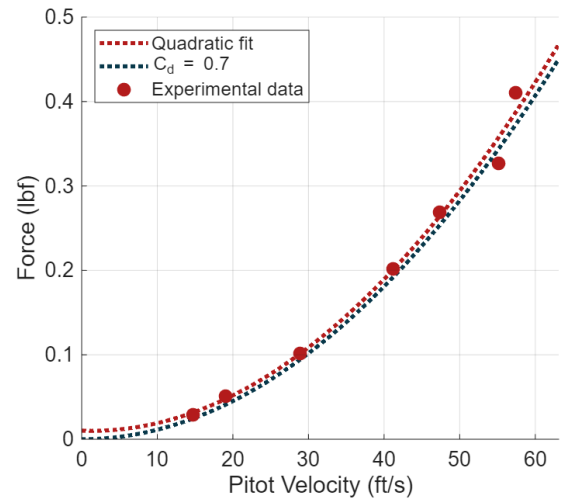


Figure 8.1.4: Nosecone windtunnel drag

Banner Release Strength Test Results A reading of **22.36 lbf** was observed on the force gauge when the piece separated from the spar. This provides a FoS of **1.49** against the expected loads of the banner, even in the worst case loading scenario.

Material	Force (lbf)		
	Trial 1	Trial 2	Trial 3
Ripstop Nylon	0.38	0.33	0.31
LDPE	0.64	0.68	0.62

Table 8.1.3: Drag forces of banner materials

8.2 Field Testing

The first flight test was conducted on November 8, 2025, with the primary objective of establishing a performance baseline and identifying areas for improvement. Prior to flight, the battery was positioned towards the rear of the fuselage and the expected CG location was confirmed. Iteration 1 achieved a successful takeoff, as shown in **Figure 8.2.1**, and completed one lap before losing transmitter connection during a left turn away from the pilot. The aircraft entered a controlled glide and impacted the ground, sustaining damage to the landing gear and propeller, shown in **Figure 8.2.1**. The stable glide demonstrated favorable aerodynamic performance, and post-flight evaluation

confirmed that the aircraft was repairable. Post-flight assessment also revealed that the receiver antenna were only affixed to one side of the fuselage, resulting in the loss of radio connection, highlighting the necessity of distributing receiver leads on both sides of the fuselage and utilizing an externally mounted receiver away from signal-blocking carbon fiber.



Figure 8.2.1: Iteration 1 successful takeoff (left) and condition after crash (right)

Flight Test 2 with Iteration 1.5 was conducted on November 14, 2025. Based on the stable glide behavior observed previously, the primary aerodynamic configuration was retained. Repairs were made to the damaged control surfaces and wing ribs, and an external receiver was added on the opposite side of the fuselage and connected to the primary receiver to reduce carbon fiber signal attenuation and improve control continuity. A CG confirmation test was again performed prior to flight, with the battery positioned accordingly. This test resulted in a successful takeoff, sustained flight, and controlled landing. The aircraft demonstrated robust maneuverability, including loops, barrel rolls, and stable operations across a range of speeds and altitudes, as shown in **Figure 8.2.2**. While Iteration 1.5 was considered a successful flight test, several design limitations were identified. The CG configuration depended on ballast and rearward battery placement, which are not feasible under competition constraints. This informed the current design documented in this report. Further flight testing was not possible due to the adverse weather conditions of Ithaca winter, and a flight test involving the banner was delayed until early March when the flight field is expected to be clear of snow.



Figure 8.2.2: Drone capture of Iteration 1.5 completing a barrel roll

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